Stochastic analysis of Bernoulli processes

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Abstract

These notes survey some aspects of discrete-time chaotic calculus and its applications, based on the chaos representation property for i.i.d. sequences of random variables. The topics covered include the Clark formula and predictable representation, anticipating calculus, covariance identities and functional inequalities (such as deviation and logarithmic Sobolev inequalities), and an application to option hedging in discrete time.

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Classification: 60G42, 60G50, 60G51, 60H30, 60H07.

1 Introduction

Stochastic analysis can be viewed as an infinite-dimensional version of classical analysis, developed in relation to stochastic processes.

In this survey we present a construction of the basic operators of stochastic analysis (gradient and divergence) in discrete time for Bernoulli processes. Our presentation is based on the chaos representation property and discrete multiple stochastic integrals with respect to i.i.d. sequences of random variables. The main applications presented are to functional inequalities (deviation inequalities, logarithmic Sobolev inequalities) in discrete settings, cf. [10], [16], [23], and to option pricing and hedging in discrete time mathematical finance.

Other approaches to discrete-time stochastic analysis can be found in Holden et al. [13] (1992), [14] (1993), Leitz-Martini [22] (2000), and also in Attal [2] (2003) in the framework of quantum stochastic calculus, see also the recent paper [12] by H. Gzyl (2005).

This survey can be roughly divided into a first part (Sections 2 to 11) in which we present the main basic results and analytic tools, and a second part (Sections 12 to 15) which is devoted to applications.

We proceed as follows. In Section 2 we consider a family of discrete-time normal martingales. The next section is devoted to the construction of the stochastic integral of predictable square-integrable processes with respect to such martingales. In Section 4 we construct the associated multiple stochastic integrals of symmetric functions on $\mathbb{N}^n$, $n \geq 1$. Starting with Section 5 we focus on a particular class of normal martingales satisfying a structure equation. The chaos representation property is studied in Section 6 in the case of discrete time random walks with independent increments. A gradient operator D acting by finite differences is introduced in Section 7 in connection with multiple stochastic integrals, and used in Section 8 to state a Clark predictable representation formula. The divergence operator δ, adjoint of D, is presented in Section 9 as an extension of the discrete-time stochastic integral. It is also used in Section 10 to express the generator of the Ornstein-Uhlenbeck process. Covariance identities are stated in Section 11, both from the Clark representation formula and by use of the Ornstein-Uhlenbeck semigroup.

Functional inequalities on Bernoulli space are presented as an application in Sections 12 and 13. On the one hand, in Section 12 we prove several deviation inequalities for functionals of an infinite number of i.i.d. Bernoulli random variables. Then in Section 13 we state different versions of the logarithmic Sobolev inequality in discrete settings (modified, L^1, sharp) which allow one to control the entropy of random variables. In particular we recover and extend some results of [5], using the method of [10]. Our approach is based on the intrinsic tools (gradient, divergence, Laplacian) of infinite-dimensional stochastic analysis. We refer to [4], [3], [17], [20], for other versions of logarithmic Sobolev inequalities in discrete settings, and to [7], [28] for the Poisson case.

Section 14 contains a change of variable formula in discrete time, which is applied with the Clark formula in Section 15 to a derivation of the Black-Scholes formula in

discrete time, i.e. in the Cox-Ross-Rubinstein model, see e.g. [19], §15-1 of [27], or [24], for other approaches.

2 Discrete-Time Normal Martingales

Consider a sequence $(Y_k)_{k\in\mathbb{N}}$ of (not necessarily independent) random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $(\mathcal{F}_n)_{n\geq -1}$ denote the filtration generated by $(Y_n)_{n\in\mathbb{N}}$, i.e.

$$\mathcal{F}_{-1} = \{\emptyset, \Omega\},$$

and

$$\mathcal{F}_n = \sigma(Y_0, \ldots, Y_n), \qquad n \geq 0.$$

Recall that a random variable F is said to be $\mathcal{F}_n$-measurable if it can be written as a function

$$F = f_n(Y_0, \ldots, Y_n)$$

of $Y_0, \ldots, Y_n$, where $f_n : \mathbb{R}^{n+1} \to \mathbb{R}$.

Assumption 2.1 *We make the following assumptions on the sequence* $(Y_n)_{n\in\mathbb{N}}$*:*

a) it is conditionally centered:

$$\mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] = 0, \qquad n \geq 0, \tag{2.1}$$

b) its conditional quadratic variation satisfies:

$$\mathbb{E}[Y_n^2 \mid \mathcal{F}_{n-1}] = 1, \qquad n \geq 0.$$

Condition (2.1) implies that the process $(Y_0 + \cdots + Y_n)_{n\geq 0}$ is an $\mathcal{F}_n$-martingale. More precisely, the sequence $(Y_n)_{n\in\mathbb{N}}$ and the process $(Y_0 + \cdots + Y_n)_{n\geq 0}$ can be viewed respectively as a (correlated) noise and as a normal martingale in discrete time.

3 Discrete Stochastic Integrals

In this section we construct the discrete stochastic integral of predictable square-summable processes with respect to a discrete-time normal martingale.

Definition 3.1 *Let $(u_k)_{k\in\mathbb{N}}$ be a uniformly bounded sequence of random variables with finite support in $\mathbb{N}$, i.e. there exists $N \geq 0$ such that $u_k = 0$ for all $k \geq N$. The stochastic integral $J(u)$ of $(u_n)_{n\in\mathbb{N}}$ is defined as*

$$J(u) = \sum_{k=0}^{\infty} u_k Y_k.$$

The next proposition states a version of the Itô isometry in discrete time. A sequence $(u_n)_{n\in\mathbb{N}}$ of random variables is said to be $\mathcal{F}_n$-predictable if u_n is $\mathcal{F}_{n-1}$-measurable for all $n \in \mathbb{N}$, in particular u_0 is constant in this case.

Proposition 3.2 *The stochastic integral operator $J(u)$ extends to square-integrable predictable processes $(u_n)_{n\in\mathbb{N}} \in L^2(\Omega \times \mathbb{N})$ via the (conditional) isometry formula*

$$\mathbb{E}[|J(\mathbf{1}_{[n,\infty)}u)|^2| \mid \mathcal{F}_{n-1}] = \mathbb{E}[\|\mathbf{1}_{[n,\infty)}u\|^2_{\ell^2(\mathbb{N})} \mid \mathcal{F}_{n-1}], \qquad n \in \mathbb{N}. \tag{3.3}$$

Proof. Let $(u_n)_{n\in\mathbb{N}}$ and $(v_n)_{n\in\mathbb{N}}$ be bounded predictable processes with finite support in $\mathbb{N}$. The product $u_k Y_k v_l$, $0 \leq k < l$, is $\mathcal{F}_{l-1}$-measurable, and $u_k Y_l v_l$ is $\mathcal{F}_{k-1}$-measurable, $0 \leq l < k$. Hence

$$\begin{aligned}
\mathbb{E}\left[\sum_{k=n}^{\infty} u_k Y_k \sum_{l=0}^{\infty} v_l Y_l \Big| \mathcal{F}_{n-1}\right] &= \mathbb{E}\left[\sum_{k,l=n}^{\infty} u_k Y_k v_l Y_l \Big| \mathcal{F}_{n-1}\right] \\
&= \mathbb{E}\left[\sum_{k=n}^{\infty} u_k v_k Y_k^2 + \sum_{n\leq k<l} u_k Y_k v_l Y_l + \sum_{n\leq l<k} u_k Y_k v_l Y_l \Big| \mathcal{F}_{n-1}\right] \\
&= \sum_{k=n}^{\infty} \mathbb{E}[\mathbb{E}[u_k v_k Y_k^2 \mid \mathcal{F}_{k-1}] \mid \mathcal{F}_{n-1}] + \sum_{n\leq k<l} \mathbb{E}[\mathbb{E}[u_k Y_k v_l Y_l \mid \mathcal{F}_{l-1}] \mid \mathcal{F}_{n-1}] \\
&\quad + \sum_{n\leq l<k} \mathbb{E}[\mathbb{E}[u_k Y_k v_l Y_l \mid \mathcal{F}_{k-1}] \mid \mathcal{F}_{n-1}] \\
&= \sum_{k=0}^{\infty} \mathbb{E}[u_k v_k \mathbb{E}[Y_k^2 \mid \mathcal{F}_{k-1}] \mid \mathcal{F}_{n-1}] + 2\sum_{n\leq k<l} \mathbb{E}[u_k Y_k v_l \mathbb{E}[Y_l \mid \mathcal{F}_{l-1}] \mid \mathcal{F}_{n-1}] \\
&= \sum_{k=n}^{\infty} \mathbb{E}[u_k v_k \mid \mathcal{F}_{n-1}] \\
&= \mathbb{E}\left[\sum_{k=n}^{\infty} u_k v_k \Big| \mathcal{F}_{n-1}\right].
\end{aligned}$$

This proves the isometry property (3.3) for J. The extension to $L^2(\Omega \times \mathbb{N})$ follows then from a Cauchy sequence argument. Consider a sequence of bounded predictable

processes with finite support converging to u in $L^2(\Omega\times\mathbb{N})$, for example the sequence $(u^n)_{n\in\mathbb{N}}$ defined as

$$u^n = (u_k^n)_{k\in\mathbb{N}} = (u_k \mathbf{1}_{\{0\leq k\leq n\}}\mathbf{1}_{\{|u_k|\leq n\}})_{k\in\mathbb{N}}, \qquad n\in\mathbb{N}.$$

Then the sequence $(J(u^n))_{n\in\mathbb{N}}$ is Cauchy and converges in $L^2(\Omega)$, hence we may define

$$J(u) := \lim_{k\to\infty} J(u^k).$$

From the isometry property (3.3) applied with $n=0$, the limit is clearly independent of the choice of the approximating sequence $(u^k)_{k\in\mathbb{N}}$. □

Note that by bilinearity, (3.3) can also be written as

$$\mathbb{E}[J(\mathbf{1}_{[n,\infty)}u)J(\mathbf{1}_{[n,\infty)}v)|\mathcal{F}_{n-1}] = \mathbb{E}[\langle \mathbf{1}_{[n,\infty)}u, \mathbf{1}_{[n,\infty)}v\rangle_{\ell^2(\mathbb{N})} \mid \mathcal{F}_{n-1}], \qquad n\in\mathbb{N},$$

and that for $n=0$ we get

$$\mathbb{E}[J(u)J(v)] = \mathbb{E}[\langle u, v\rangle_{\ell^2(\mathbb{N})}], \tag{3.4}$$

for all square-integrable predictable processes $u=(u_k)_{k\in\mathbb{N}}$ and $v=(v_k)_{k\in\mathbb{N}}$.

Proposition 3.5 *Let $(u_k)_{k\in\mathbb{N}}\in L^2(\Omega\times\mathbb{N})$ be a predictable square-integrable process. We have*

$$\mathbb{E}[J(u)\mid\mathcal{F}_k] = J(u\mathbf{1}_{[0,k]}), \quad k\in\mathbb{N}.$$

Proof. It is sufficient to note that

$$\begin{aligned}
\mathbb{E}[J(u)\mid\mathcal{F}_k] &= \mathbb{E}\left[\sum_{i=0}^{k} u_iY_i\Big|\mathcal{F}_k\right] + \sum_{i=k+1}^{\infty}\mathbb{E}\left[u_iY_i\mid\mathcal{F}_k\right] \\
&= \sum_{i=0}^{k} u_iY_i + \sum_{i=k+1}^{\infty}\mathbb{E}\left[\mathbb{E}\left[u_iY_i\mid\mathcal{F}_{i-1}\right]\mid\mathcal{F}_k\right] \\
&= \sum_{i=0}^{k} u_iY_i + \sum_{i=k+1}^{\infty}\mathbb{E}\left[u_i\mathbb{E}\left[Y_i\mid\mathcal{F}_{i-1}\right]\mid\mathcal{F}_k\right] \\
&= \sum_{i=0}^{k} u_iY_i \\
&= J(u\mathbf{1}_{[0,k]}).
\end{aligned}$$

□

Corollary 3.6 *The indefinite stochastic integral* $(J(u\mathbf{1}_{[0,k]}))_{k\in\mathbb{N}}$ *is a discrete time martingale with respect to* $(\mathcal{F}_n)_{n\geq -1}$.

Proof. We have

$$\begin{aligned}\mathbb{E}[J(u\mathbf{1}_{[0,k+1]}) \mid \mathcal{F}_k] &= \mathbb{E}[\mathbb{E}[J(u\mathbf{1}_{[0,k+1]}) \mid \mathcal{F}_{k+1} \mid \mathcal{F}_k] \\ &= \mathbb{E}[\mathbb{E}[J(u) \mid \mathcal{F}_{k+1} \mid \mathcal{F}_k] \\ &= \mathbb{E}[J(u) \mid \mathcal{F}_k] \\ &= J(u\mathbf{1}_{[0,k]}).\end{aligned}$$

□

4 Discrete Multiple Stochastic Integrals

The role of multiple stochastic integrals in the orthogonal expansions of random variables is similar to that of polynomials in the series expansions of functions of a real variable. In some cases, multiple stochastic integrals can be expressed using polynomials, for example Krawtchouk polynomials in the symmetric discrete case with $p_n = q_n = 1/2$, $n \in \mathbb{N}$, see Relation (6.2) below.

Definition 4.1 *Let* $\ell^2(\mathbb{N})^{\circ n}$ *denote the subspace of* $\ell^2(\mathbb{N})^{\otimes n} = \ell^2(\mathbb{N}^n)$ *made of functions* f_n *that are symmetric in* n *variables, i.e. such that for every permutation* σ *of* $\{1,\ldots,n\}$,

$$f_n(k_{\sigma(1)},\ldots,k_{\sigma(n)}) = f_n(k_1,\ldots,k_n), \quad k_1,\ldots,k_n \in \mathbb{N}.$$

Given $f_1 \in l^2(\mathbb{N})$ we let

$$J_1(f_1) = J(f_1) = \sum_{k=0}^{\infty} f_1(k)Y_k.$$

As a convention we identify $\ell^2(\mathbb{N}^0)$ to $\mathbb{R}$ and let $J_0(f_0) = f_0$, $f_0 \in \mathbb{R}$. Let

$$\Delta_n = \{(k_1,\ldots,k_n) \in \mathbb{N}^n \ : \ k_i \neq k_j, \ 1 \leq i < j \leq n\}, \qquad n \geq 1.$$

The following proposition gives the definition of multiple stochastic integrals by iterated stochastic integration of predictable processes in the sense of Proposition 3.2.

Proposition 4.2 *The multiple stochastic integral $J_n(f_n)$ of $f_n \in \ell^2(\mathbb{N})^{\circ n}$, $n \geq 1$, is defined as*

$$J_n(f_n) = \sum_{(i_1,\ldots,i_n)\in\Delta_n} f_n(i_1,\ldots,i_n) Y_{i_1} \cdots Y_{i_n}.$$

It satisfies the recurrence relation

$$J_n(f_n) = n \sum_{k=1}^{\infty} Y_k J_{n-1}(f_n(*,k)\mathbf{1}_{[0,k-1]^{n-1}}(*)) \tag{4.3}$$

and the isometry formula

$$\mathbb{E}[J_n(f_n)J_m(g_m)] = \begin{cases} n!\langle \mathbf{1}_{\Delta_n} f_n, g_m\rangle_{\ell^2(\mathbb{N})^{\otimes n}} & \text{if } n = m, \\ 0 & \text{if } n \neq m. \end{cases} \tag{4.4}$$

Proof. Note that we have

$$\begin{aligned} J_n(f_n) &= n! \sum_{0\leq i_1<\cdots<i_n} f_n(i_1,\ldots,i_n) Y_{i_1}\cdots Y_{i_n} \\ &= n! \sum_{i_n=0}^{\infty} \sum_{0\leq i_{n-1}<i_n} \cdots \sum_{0\leq i_1<i_2} f_n(i_1,\ldots,i_n) Y_{i_1}\cdots Y_{i_n}. \end{aligned} \tag{4.5}$$

Note that since $0 \leq i_1 < i_2 < \cdots < i_n$ and $0 \leq j_1 < j_2 < \cdots < j_n$ we have

$$\mathbb{E}[Y_{i_1}\cdots Y_{i_n} Y_{j_1}\cdots Y_{j_n}] = \mathbf{1}_{\{i_1=j_1,\ldots,i_n=j_n\}}.$$

Hence

$$\begin{aligned} &\mathbb{E}[J_n(f_n)J_n(g_n)] \\ &= (n!)^2 \mathbb{E}\left[\sum_{0\leq i_1<\cdots<i_n} f_n(i_1,\ldots,i_n) Y_{i_1}\cdots Y_{i_n} \sum_{0\leq j_1<\cdots<j_n} g_n(j_1,\ldots,j_n) Y_{j_1}\cdots Y_{j_n} \right] \\ &= (n!)^2 \sum_{0\leq i_1<\cdots<i_n,\ 0\leq j_1<\cdots<j_n} f_n(i_1,\ldots,i_n) g_n(j_1,\ldots,j_n) \mathbb{E}[Y_{i_1}\cdots Y_{i_n} Y_{j_1}\cdots Y_{j_n}] \\ &= (n!)^2 \sum_{0\leq i_1<\cdots<i_n} f_n(i_1,\ldots,i_n) g_n(i_1,\ldots,i_n) \\ &= n! \sum_{(i_1,\ldots,i_n)\in\Delta_n} f_n(i_1,\ldots,i_n) g_n(i_1,\ldots,i_n) \\ &= n! \langle \mathbf{1}_{\Delta_n} f_n, g_m\rangle_{\ell^2(\mathbb{N})^{\otimes n}}. \end{aligned}$$

When $n < m$ and $(i_1,\ldots,i_n) \in \Delta_n$ and $(j_1,\ldots,j_m) \in \Delta_m$ are two sets of indices, there necessarily exists $k \in \{1,\ldots,m\}$ such that $j_k \notin \{i_1,\ldots,i_n\}$, hence

$$\mathbb{E}[Y_{i_1}\cdots Y_{i_n} Y_{j_1}\cdots Y_{j_m}] = 0,$$

and this implies the orthogonality of $J_n(f_n)$ and $J_m(g_m)$. The recurrence relation (4.3) is a direct consequence of (4.5). The isometry property (4.4) of J_n also follows by induction from (3.3) and the recurrence relation. □

If $f_n \in \ell^2(\mathbb{N}^n)$ is not symmetric we let $J_n(f_n) = J_n(\tilde{f}_n)$, where $\tilde{f}_n$ is the symmetrization of f_n, defined as

$$\tilde{f}_n(i_1, \ldots, i_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} f(i_{\sigma(1)}, \ldots, i_{\sigma_n}), \qquad i_1, \ldots, i_n \in \mathbb{N}^n,$$

and Σ_n is the set of all permutations of $\{1, \ldots, n\}$. In particular, if $(k_1, \ldots, k_n) \in \Delta_n$, the symmetrization $\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_n)\}}$ of $\mathbf{1}_{\{(k_1,\ldots,k_n)\}}$ in n variables is given by

$$\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_n)\}}(i_1, \ldots, i_n) = \frac{1}{n!} \mathbf{1}_{\{\{i_1,\ldots,i_n\}=\{k_1,\ldots,k_n\}\}}, \quad i_1, \ldots, i_n \in \mathbb{N},$$

and

$$J_n(\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_n)\}}) = Y_{k_1} \cdots Y_{k_n}.$$

Lemma 4.6 *For all $n \geq 1$ we have*

$$\mathbb{E}[J_n(f_n) \mid \mathcal{F}_k] = J_n(f_n \mathbf{1}_{[0,k]^n}), \qquad k \in \mathbb{N}, \quad f_n \in \ell^2(\mathbb{N})^{\circ n}.$$

Proof. This lemma can be proved in two ways, either as a consequence of Proposition 3.5 and Proposition 4.2 or via the following direct argument, noting that for all $m = 0, \ldots, n$ and $g_m \in \ell^2(\mathbb{N})^{\circ m}$ we have:

$$\begin{aligned} \mathbb{E}[(J_n(f_n) - J_n(f_n \mathbf{1}_{[0,k]^n})) J_m(g_m \mathbf{1}_{[0,k]^m})] &= \mathbf{1}_{\{n=m\}} n! \langle f_n(1 - \mathbf{1}_{[0,k]^n}), g_m \mathbf{1}_{[0,k]^m} \rangle_{\ell^2(\mathbb{N}^n)} \\ &= 0, \end{aligned}$$

hence $J_n(f_n \mathbf{1}_{[0,k]^n}) \in L^2(\Omega, \mathcal{F}_k)$, and $J_n(f_n) - J_n(f_n \mathbf{1}_{[0,k]^n})$ is orthogonal to $L^2(\Omega, \mathcal{F}_k)$. □

In other terms we have

$$\mathbb{E}[J_n(f_n)] = 0, \qquad f_n \in \ell^2(\mathbb{N})^{\circ n}, \qquad n \geq 1,$$

the process $(J_n(f_n \mathbf{1}_{[0,k]^n}))_{k \in \mathbb{N}}$ is a discrete-time martingale, and $J_n(f_n)$ is $\mathcal{F}_k$-measurable if and only if $f_n \mathbf{1}_{[0,k]^n} = f_n$, $0 \leq k \leq n$.

5 Discrete structure equations

Assume now that the sequence $(Y_n)_{n \in \mathbb{N}}$ satisfies the discrete structure equation:

$$Y_n^2 = 1 + \varphi_n Y_n, \qquad n \in \mathbb{N}, \tag{5.1}$$

where $(\varphi_n)_{n\in\mathbb{N}}$ is an $\mathcal{F}_n$-predictable process. Condition (2.1) implies that

$$\mathbb{E}[Y_n^2 \mid \mathcal{F}_{n-1}] = 1, \qquad n \in \mathbb{N},$$

hence the hypotheses of the preceding sections are satisfied. Since (5.1) is a second order equation, there exists an $\mathcal{F}_n$-adapted process $(X_n)_{n\in\mathbb{N}}$ of Bernoulli $\{-1,1\}$-valued random variables such that

$$Y_n = \frac{\varphi_n}{2} + X_n\sqrt{1+\left(\frac{\varphi_n}{2}\right)^2}, \qquad n \in \mathbb{N}. \tag{5.2}$$

Consider the conditional probabilities

$$p_n = \mathbb{P}(X_n = 1 \mid \mathcal{F}_{n-1}) \quad \text{and} \quad q_n = \mathbb{P}(X_n = -1 \mid \mathcal{F}_{n-1}), \qquad n \in \mathbb{N}. \tag{5.3}$$

From the relation $\mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] = 0$, rewritten as

$$p_n\left(\frac{\varphi_n}{2} + \sqrt{1+\left(\frac{\varphi_n}{2}\right)^2}\right) + q_n\left(\frac{\varphi_n}{2} - \sqrt{1+\left(\frac{\varphi_n}{2}\right)^2}\right) = 0, \qquad n \in \mathbb{N},$$

we get

$$p_n = \frac{1}{2}\left(1 - \frac{\varphi_n}{\sqrt{4+\varphi_n^2}}\right), \qquad q_n = \frac{1}{2}\left(1 + \frac{\varphi_n}{\sqrt{4+\varphi_n^2}}\right), \tag{5.4}$$

and

$$\varphi_n = \sqrt{\frac{q_n}{p_n}} - \sqrt{\frac{p_n}{q_n}} = \frac{q_n - p_n}{\sqrt{p_n q_n}}, \qquad n \in \mathbb{N},$$

hence

$$Y_n = \mathbf{1}_{\{X_n=1\}}\sqrt{\frac{q_n}{p_n}} - \mathbf{1}_{\{X_n=-1\}}\sqrt{\frac{p_n}{q_n}}, \qquad n \in \mathbb{N}.$$

Letting

$$Z_n = \frac{X_n+1}{2} \in \{0,1\}, \qquad n \in \mathbb{N},$$

we also have the relations

$$Y_n = \frac{q_n - p_n + X_n}{2\sqrt{p_n q_n}} = \frac{Z_n - p_n}{\sqrt{p_n q_n}}, \quad n \in \mathbb{N}, \tag{5.5}$$

which yield

$$\mathcal{F}_n = \sigma(X_0,\ldots,X_n) = \sigma(Z_0,\ldots,Z_n), \qquad n \in \mathbb{N}.$$

Remark 5.6 *In particular, one can take* $\Omega = \{-1,1\}^{\mathbb{N}}$ *and construct the Bernoulli process* $(X_n)_{n\in\mathbb{N}}$ *as the sequence of canonical projections on* $\Omega = \{-1,1\}^{\mathbb{N}}$ *under*

a countable product $\mathbb{P}$ of Bernoulli measures on $\{-1,1\}$. In this case the sequence $(X_n)_{n\in\mathbb{N}}$ can be viewed as the dyadic expansion of $X(\omega)\in[0,1]$ defined as:

$$X(\omega)=\sum_{n=0}^{\infty}\frac{1}{2^{n+1}}X_n(\omega).$$

In the symmetric case $p_k=q_k=1/2$, $k\in\mathbb{N}$, the image measure of $\mathbb{P}$ by the mapping $\omega\mapsto X(\omega)$ is the Lebesgue measure on $[0,1]$, see [26] for the non-symmetric case.

6 Chaos representation

From now on we assume that the sequence $(p_k)_{k\in\mathbb{N}}$ defined in (5.3) is deterministic, which implies that the random variables $(X_n)_{n\in\mathbb{N}}$ are independent. Precisely, X_n will be constructed as the canonical projection $X_n:\Omega\to\{-1,1\}$ on $\Omega=\{-1,1\}^{\mathbb{N}}$ under the measure $\mathbb{P}$ given on cylinder sets by

$$\mathbb{P}(\{\epsilon_0,\ldots,\epsilon_n\}\times\{-1,1\}^{\mathbb{N}})=\prod_{k=0}^{n}p_k^{(1+\varepsilon_k)/2}q_k^{(1-\varepsilon_k)/2},\qquad\{\epsilon_0,\ldots,\epsilon_n\}\in\{-1,1\}^{n+1}.$$

The sequence $(Y_k)_{k\in\mathbb{N}}$ can be constructed as a family of independent random variables given by

$$Y_n=\frac{\varphi_n}{2}+X_n\sqrt{1+\left(\frac{\varphi_n}{2}\right)^2},\qquad n\in\mathbb{N},$$

where the sequence $(\varphi_n)_{n\in\mathbb{N}}$ is deterministic. In this case, all spaces $L^r(\Omega,\mathcal{F}_n)$, $r\geq 1$, have finite dimension 2^{n+1}, with basis

$$\begin{aligned}&\left\{\mathbf{1}_{\{Y_0=\epsilon_0,\ldots,Y_n=\epsilon_n\}}\ :\ (\epsilon_0,\ldots,\epsilon_n)\in\prod_{k=0}^{n}\left\{\sqrt{\frac{q_k}{p_k}},-\sqrt{\frac{p_k}{q_k}}\right\}\right\}\\&\quad=\quad\left\{\mathbf{1}_{\{X_0=\epsilon_0,\ldots,X_n=\epsilon_n\}}\ :\ (\epsilon_0,\ldots,\epsilon_n)\in\prod_{k=0}^{n}\{-1,1\}\right\}.\end{aligned}$$

An orthogonal basis of $L^r(\Omega,\mathcal{F}_n)$ is given by

$$\left\{Y_{k_1}\cdots Y_{k_l}=J_l(\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_l)\}})\ :\ 0\leq k_1<\cdots<k_l\leq n,\ l=0,\ldots,n+1\right\}.$$

Let

$$S_n=\sum_{k=0}^{n}\frac{1+X_k}{2}=\sum_{k=0}^{n}Z_k,\quad n\in\mathbb{N},\tag{6.1}$$

denote the random walk associated to $(X_k)_{k\in\mathbb{N}}$. If $p_k=p$, $k\in\mathbb{N}$, then

$$J_n(\mathbf{1}_{[0,N]}^{\circ n})=K_n(S_N;N+1,p)\tag{6.2}$$

coincides with the Krawtchouk polynomial $K_n(\cdot; N+1, p)$ of order n and parameter $(N+1, p)$, evaluated at S_N, cf. [23].

Let now $\mathcal{H}_0 = \mathbb{R}$ and let $\mathcal{H}_n$ denote the subspace of $L^2(\Omega)$ made of integrals of order $n \geq 1$, and called chaos of order n:

$$\mathcal{H}_n = \{J_n(f_n) \ : \ f_n \in \ell^2(\mathbb{N})^{\circ n}\}.$$

The space of $\mathcal{F}_n$-measurable random variables is denoted by $L^0(\Omega, \mathcal{F}_n)$.

Lemma 6.3 *For all $n \in \mathbb{N}$ we have*

$$L^0(\Omega, \mathcal{F}_n) \subset \mathcal{H}_0 \oplus \cdots \oplus \mathcal{H}_{n+1}. \tag{6.4}$$

Proof. It suffices to note that $\mathcal{H}_l \cap L^0(\Omega, \mathcal{F}_n)$ has dimension $\binom{n+1}{l}$, $1 \leq l \leq n+1$. More precisely it is generated by the orthonormal basis

$$\left\{Y_{k_1} \cdots Y_{k_l} = J_l(\widetilde{\mathbf{1}}_{\{(k_1,\ldots,k_l)\}}) \ : \ 0 \leq k_1 < \cdots < k_l \leq n\right\},$$

since any element F of $\mathcal{H}_l \cap L^0(\Omega, \mathcal{F}_n)$ can be written as $F = J_l(f_l \mathbf{1}_{[0,n]^l})$, hence

$$L^0(\Omega, \mathcal{F}_n) = (\mathcal{H}_0 \oplus \cdots \oplus \mathcal{H}_{n+1}) \bigcap L^0(\Omega, \mathcal{F}_n).$$

□

Alternatively, Lemma 6.3 can be proved by noting that

$$J_n(f_n \mathbf{1}_{[0,N]^n}) = 0, \qquad n > N+1, \quad f_n \in \ell^2(\mathbb{N})^{\circ n},$$

and as a consequence, any $F \in L^0(\Omega, \mathcal{F}_N)$ can be expressed as

$$F = \mathbb{E}[F] + \sum_{n=1}^{N+1} J_n(f_n \mathbf{1}_{[0,N]^n}).$$

Definition 6.5 *Let $\mathcal{S}$ denote the linear space spanned by multiple stochastic integrals, i.e.*

$$\mathcal{S} = \mathrm{Vect}\left\{\bigcup_{n=0}^{\infty} \mathcal{H}_n\right\} = \left\{\sum_{k=0}^{n} J_k(f_k) \ : \ f_k \in \ell^2(\mathbb{N})^{\circ k}, \ k = 0, \ldots, n, \ n \in \mathbb{N}\right\}. \tag{6.6}$$

The completion of $\mathcal{S}$ in $L^2(\Omega)$ is denoted by the direct sum

$$\bigoplus_{n=0}^{\infty} \mathcal{H}_n.$$

The next result is the chaos representation property for Bernoulli processes, which is analogous to the Walsh decomposition, cf. [22]. This property is obtained under the assumption that the sequence $(X_n)_{n\in\mathbb{N}}$ is i.i.d. See [8] for other instances of the chaos representation property without this independence assumption.

Proposition 6.7 *We have the identity*

$$L^2(\Omega) = \bigoplus_{n=0}^{\infty} \mathcal{H}_n.$$

Proof. It suffices to show that $\mathcal{S}$ is dense in $L^2(\Omega)$. Let F be a bounded random variable. Relation (6.4) of Lemma 6.3 shows that $\mathbb{E}[F \mid \mathcal{F}_n] \in \mathcal{S}$. The martingale convergence theorem, cf. e.g. Theorem 27.1 in [18], implies that $(\mathbb{E}[F \mid \mathcal{F}_n])_{n\in\mathbb{N}}$ converges to F a.s., hence every bounded F is the $L^2(\Omega)$-limit of a sequence in $\mathcal{S}$. If $F \in L^2(\Omega)$ is not bounded, F is the limit in $L^2(\Omega)$ of the sequence $(\mathbf{1}_{\{|F|\leq n\}}F)_{n\in\mathbb{N}}$ of bounded random variables. □

As a consequence of Proposition 6.7, any $F \in L^2(\Omega, \mathbb{P})$ has a unique decomposition

$$F = \mathbb{E}[F] + \sum_{n=1}^{\infty} J_n(f_n), \qquad f_n \in l^2(\mathbb{N})^{\circ n},\ n \in \mathbb{N},$$

as a series of multiple stochastic integrals. Note also that the statement of Lemma 6.3 is sufficient for the chaos representation property to hold.

7 Gradient Operator

Definition 7.1 *We densely define the linear gradient operator*

$$D : \mathcal{S} \longrightarrow L^2(\Omega \times \mathbb{N})$$

by

$$D_k J_n(f_n) = n J_{n-1}(f_n(*, k)\mathbf{1}_{\Delta_n}(*, k)),$$

$k \in \mathbb{N}$, $f_n \in \ell^2(\mathbb{N})^{\circ n}$ $n \in \mathbb{N}$.

Note that for all $k_1, \ldots, k_{n-1}, k \in \mathbb{N}$ we have

$$\mathbf{1}_{\Delta_n}(k_1, \ldots, k_{n-1}, k) = \mathbf{1}_{\{k \notin (k_1, \ldots, k_{n-1})\}} \mathbf{1}_{\Delta_{n-1}}(k_1, \ldots, k_{n-1}),$$

hence we can write

$$D_k J_n(f_n) = n J_{n-1}(f_n(*, k) \mathbf{1}_{\{k \notin *\}}), \quad k \in \mathbb{N},$$

where in the above relation, "$*$" denotes the first $k-1$ variables $(k_1, \ldots, k_{n-1})$ of $f_n(k_1, \ldots, k_{n-1}, k)$. We also have $D_k F = 0$ whenever $F \in \mathcal{S}$ is $\mathcal{F}_{k-1}$-measurable.

On the other hand, D_k is a continuous operator on the chaos $\mathcal{H}_n$ since

$$\begin{aligned} \|D_k J_n(f_n)\|^2_{L^2(\Omega)} &= n^2 \|J_{n-1}(f_n(*, k))\|^2_{L^2(\Omega)} \\ &= nn! \|f_n(*, k)\|^2_{\ell^2(\mathbb{N}^{\otimes(n-1)})}, \qquad f_n \in \ell^2(\mathbb{N}^{\otimes n}), \quad k \in \mathbb{N}. \end{aligned} \tag{7.2}$$

The following result gives the probabilistic interpretation of D_k as a finite difference operator. Given

$$\omega = (\omega_0, \omega_1, \ldots) \in \{-1, 1\}^{\mathbb{N}},$$

let

$$\omega^k_+ = (\omega_0, \omega_1, \ldots, \omega_{k-1}, +1, \omega_{k+1}, \ldots)$$

and

$$\omega^k_- = (\omega_0, \omega_1, \ldots, \omega_{k-1}, -1, \omega_{k+1}, \ldots).$$

Proposition 7.3 *We have for any $F \in \mathcal{S}$:*

$$D_k F(\omega) = \sqrt{p_k q_k}(F(\omega^k_+) - F(\omega^k_-)), \quad k \in \mathbb{N}. \tag{7.4}$$

Proof. We start by proving the above statement for an $\mathcal{F}_n$-measurable $F \in \mathcal{S}$. Since $L^0(\Omega, \mathcal{F}_n)$ is finite dimensional it suffices to consider

$$F = Y_{k_1} \cdots Y_{k_l} = f(X_0, \ldots, X_{k_l}),$$

with from (5.5):

$$f(x_0, \ldots, x_{k_l}) = \frac{1}{2^l} \prod_{i=1}^{l} \frac{q_{k_i} - p_{k_i} + x_{k_i}}{\sqrt{p_{k_i} q_{k_i}}}.$$

First we note that from (6.4) we have for $(k_1, \ldots, k_n) \in \Delta_n$:

$$D_k \left(Y_{k_1} \cdots Y_{k_n} \right) = D_k J_n(\tilde{\mathbf{1}}_{\{(k_1, \ldots, k_n)\}})$$

$$
\begin{aligned}
&= nJ_{n-1}(\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_n)\}}(*,k)) \\
&= \frac{1}{(n-1)!}\sum_{i=1}^{n}\mathbf{1}_{\{k_i\}}(k)\sum_{(i_1,\ldots,i_{n-1})\in\Delta_{n-1}}\tilde{\mathbf{1}}_{\{\{i_1,\ldots,i_{n-1}\}=\{k_1,\ldots,k_{i-1},k_{i+1},\ldots,k_n\}\}} \\
&= \sum_{i=1}^{n}\mathbf{1}_{\{k_i\}}(k)J_{n-1}(\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_{i-1},k_{i+1},\ldots,k_n)\}}) \\
&= \mathbf{1}_{\{k_1,\ldots,k_n\}}(k)\prod_{\substack{i=1\\k_i\neq k}}^{n}Y_{k_i}. \qquad (7.5)
\end{aligned}
$$

If $k\notin\{k_1,\ldots,k_l\}$ we clearly have $F(\omega_+^k)=F(\omega_-^k)=F(\omega)$, hence

$$\sqrt{p_kq_k}(F(\omega_+^k)-F(\omega_-^k))=0=D_kF(\omega).$$

On the other hand if $k\in\{k_1,\ldots,k_l\}$ we have

$$F(\omega_+^k)=\sqrt{\frac{q_k}{p_k}}\prod_{\substack{i=1\\k_i\neq k}}^{l}\frac{q_{k_i}-p_{k_i}+\omega_{k_i}}{2\sqrt{p_{k_i}q_{k_i}}},$$

$$F(\omega_-^k)=-\sqrt{\frac{p_k}{q_k}}\prod_{\substack{i=1\\k_i\neq k}}^{l}\frac{q_{k_i}-p_{k_i}+\omega_{k_i}}{2\sqrt{p_{k_i}q_{k_i}}},$$

hence from (7.5) we get

$$
\begin{aligned}
\sqrt{p_kq_k}(F(\omega_+^k)-F(\omega_-^k)) &= \frac{1}{2^{l-1}}\prod_{\substack{i=1\\k_i\neq k}}^{l}\frac{q_{k_i}-p_{k_i}+\omega_{k_i}}{\sqrt{p_{k_i}q_{k_i}}} \\
&= \prod_{\substack{i=1\\k_i\neq k}}^{l}Y_{k_i}(\omega) \\
&= D_k\left(Y_{k_1}\cdots Y_{k_l}\right)(\omega) \\
&= D_kF(\omega).
\end{aligned}
$$

In the general case, $J_l(f_l)$ is the L^2-limit of the sequence $\mathbb{E}[J_l(f_l)\mid\mathcal{F}_n]=J_l(f_l\mathbf{1}_{[0,n]^l})$ as n goes to infinity, and since from (7.2) the operator D_k is continuous on all chaoses $\mathcal{H}_n$, $n\geq 1$, we have

$$
\begin{aligned}
D_kF &= \lim_{n\to\infty}D_k\mathbb{E}[F\mid\mathcal{F}_n] \\
&= \lim_{n\to\infty}(\mathbb{E}[F\mid\mathcal{F}_n](\omega_+^k)-\mathbb{E}[F\mid\mathcal{F}_n](\omega_-^k)) \\
&= \sqrt{p_kq_k}(F(\omega_+^k)-F(\omega_-^k)),\quad k\in\mathbb{N}.
\end{aligned}
$$

□

The next property follows immediately from Proposition 7.3 .

Corollary 7.6 *A random variable $F : \Omega \to \mathbb{R}$ is $\mathcal{F}_n$-measurable if and only if*

$$D_k F = 0$$

for all $k > n$.

If F has the form $F = f(X_0, \ldots, X_n)$, we may also write

$$D_k F = \sqrt{p_k q_k}(F_k^+ - F_k^-), \qquad k \in \mathbb{N},$$

with

$$F_k^+ = f(X_0, \ldots, X_{k-1}, +1, X_{k+1}, \ldots, X_n),$$

and

$$F_k^- = f(X_0, \ldots, X_{k-1}, -1, X_{k+1}, \ldots, X_n).$$

The gradient D can also be expressed as

$$D_k F(S_\cdot) = \sqrt{p_k q_k}\left(F\left(S_\cdot + \mathbf{1}_{\{X_k=-1\}}\mathbf{1}_{\{k\leq\cdot\}}\right) - F\left(S_\cdot - \mathbf{1}_{\{X_k=1\}}\mathbf{1}_{\{k\leq\cdot\}}\right)\right),$$

where $F(S_\cdot)$ is an informal notation for the random variable F estimated on a given path of $(S_n)_{n\in\mathbb{N}}$ defined in (6.1) and $S_\cdot + \mathbf{1}_{\{X_k=\mp 1\}}\mathbf{1}_{\{k\leq\cdot\}}$ denotes the path of $(S_n)_{n\in\mathbb{N}}$ perturbed by forcing X_k to be equal to ± 1.

We will also use the gradient ∇_k defined as

$$\nabla_k F = X_k\left(f(X_0, \ldots, X_{k-1}, -1, X_{k+1}, \ldots, X_n) - f(X_0, \ldots, X_{k-1}, 1, X_{k+1}, \ldots, X_n)\right), \tag{7.7}$$

$k \in \mathbb{N}$, with the relation

$$D_k = -X_k\sqrt{p_k q_k}\nabla_k, \qquad k \in \mathbb{N},$$

showing that $|D_k F|^2$ coincides with $p_k q_k |\nabla_k F|^2$. From now on, D_k denotes the finite difference operator which is extended to any $F : \Omega \to \mathbb{R}$ using Relation (7.4). The L^2-domain of D is naturally defined as the space of functionals F such that $\mathbb{E}[\|DF\|^2_{\ell^2(\mathbb{N})}] < \infty$, or equivalently

$$\sum_{n=0}^{\infty} n n! \|f_n\|^2_{\ell^2(\mathbb{N}^n)} < \infty,$$

if $F = \sum_{n=0}^{\infty} J_n(f_n)$. The following is the product rule for the operator D.

Proposition 7.8 *Let $F, G : \Omega \to \mathbb{R}$. We have*

$$D_k(FG) = FD_kG + GD_kF - \frac{X_k}{\sqrt{p_k q_k}} D_kF D_kG, \qquad k \in \mathbb{N}.$$

Proof. Let $F_+^k(\omega) = F(\omega_+^k)$, $F_-^k(\omega) = F(\omega_-^k)$, $k \geq 0$. We have

$$\begin{aligned} D_k(FG) &= \sqrt{p_k q_k}(F_+^k G_+^k - F_-^k G_-^k) \\ &= \mathbf{1}_{\{X_k=-1\}}\sqrt{p_k q_k}\left(F(G_+^k - G) + G(F_+^k - F) + (F_+^k - F)(G_+^k - G)\right) \\ &\quad + \mathbf{1}_{\{X_k=1\}}\sqrt{p_k q_k}\left(F(G - G_-^k) + G(F - F_-^k) - (F - F_-^k)(G - G_-^k)\right) \\ &= \mathbf{1}_{\{X_k=-1\}}\left(FD_kG + GD_kF + \frac{1}{\sqrt{p_k q_k}} D_kF D_kG\right) \\ &\quad + \mathbf{1}_{\{X_k=1\}}\left(FD_kG + GD_kF - \frac{1}{\sqrt{p_k q_k}} D_kF D_kG\right). \end{aligned}$$

□

8 Clark Formula and Predictable Representation

In this section we prove a predictable representation formula for the functionals of $(S_n)_{n\geq 0}$ defined in (6.1).

Proposition 8.1 *For all $F \in \mathcal{S}$ we have*

$$\begin{aligned} F &= \mathbb{E}[F] + \sum_{k=0}^{\infty} \mathbb{E}[D_kF \mid \mathcal{F}_{k-1}]Y_k \qquad (8.2) \\ &= \mathbb{E}[F] + \sum_{k=0}^{\infty} Y_k D_k \mathbb{E}[F \mid \mathcal{F}_k]. \end{aligned}$$

Proof. The formula is obviously true for $F = J_0(f_0)$. Given $n \geq 1$, as a consequence of Proposition 4.2 above and Lemma 4.6 we have:

$$\begin{aligned} J_n(f_n) &= n\sum_{k=0}^{\infty} J_{n-1}(f_n(*,k)\mathbf{1}_{[0,k-1]^{n-1}}(*))Y_k \\ &= n\sum_{k=0}^{\infty} J_{n-1}(f_n(*,k)\mathbf{1}_{\Delta_n}(*,k)\mathbf{1}_{[0,k-1]^{n-1}}(*))Y_k \\ &= n\sum_{k=0}^{\infty} \mathbb{E}[J_{n-1}(f_n(*,k)\mathbf{1}_{\Delta_n}(*,k)) \mid \mathcal{F}_{k-1}]Y_k \\ &= \sum_{k=0}^{\infty} \mathbb{E}[D_k J_n(f_n) \mid \mathcal{F}_{k-1}]Y_k, \end{aligned}$$

which yields (8.2) for $F = J_n(f_n)$, since $\mathbb{E}[J_n(f_n)] = 0$. By linearity the formula is established for $F \in \mathcal{S}$. □

Although the operator D is unbounded we have the following result, which states the boundedness of the operator that maps a random variable to the unique process involved in its predictable representation.

Lemma 8.3 *The operator*

$$\begin{aligned} L^2(\Omega) &\longrightarrow L^2(\Omega \times \mathbb{N}) \\ F &\mapsto (\mathbb{E}[D_k F \mid \mathcal{F}_{k-1}])_{k \in \mathbb{N}} \end{aligned}$$

is bounded with norm equal to one.

Proof. Let $F \in \mathcal{S}$. From Relation (8.2) and the isometry formula (3.4) for the stochastic integral operator J we get

$$\begin{aligned} \|\mathbb{E}[D_\cdot F \mid \mathcal{F}_{\cdot -1}]\|^2_{L^2(\Omega \times \mathbb{N})} &= \|F - \mathbb{E}[F]\|^2_{L^2(\Omega)} \\ &\leq \|F - \mathbb{E}[F]\|^2_{L^2(\Omega)} + (\mathbb{E}[F])^2 \\ &= \|F\|^2_{L^2(\Omega)}, \end{aligned} \tag{8.4}$$

with equality in case $F = J_1(f_1)$. □

As a consequence of Lemma 8.3 we have the following corollary.

Corollary 8.5 *The Clark formula of Proposition 8.1 extends to any $F \in L^2(\Omega)$.*

Proof. Since $F \mapsto \mathbb{E}[D_\cdot F \mid \mathcal{F}_{\cdot -1}]$ is bounded from Lemma 8.3, the Clark formula extends to $F \in L^2(\Omega)$ by a standard Cauchy sequence argument. For the second identity we use the relation

$$\mathbb{E}[D_k F \mid \mathcal{F}_{k-1}] = D_k \mathbb{E}[F \mid \mathcal{F}_k]$$

which clearly holds since $D_k F$ is independent of X_k, $k \in \mathbb{N}$. □

Let us give a first elementary application of the above construction to the proof of a Poincaré inequality on Bernoulli space. We have

$$\begin{aligned} \mathrm{var}\,(F) &= \mathbb{E}[|F - \mathbb{E}[F]|^2] \\ &= \mathbb{E}\left[\left(\sum_{k=0}^{\infty} \mathbb{E}[D_k F \mid \mathcal{F}_{k-1}] Y_k \right)^2 \right] \\ &= \mathbb{E}\left[\sum_{k=0}^{\infty} (\mathbb{E}[D_k F \mid \mathcal{F}_{k-1}])^2 \right] \end{aligned}$$

$$\begin{aligned} &\leq \mathbb{E}\left[\sum_{k=0}^{\infty}\mathbb{E}[|D_kF|^2 \mid \mathcal{F}_{k-1}]\right] \\ &= \mathbb{E}\left[\sum_{k=0}^{\infty}|D_kF|^2\right], \end{aligned}$$

hence

$$\mathrm{var}\,(F) \leq \|DF\|^2_{L^2(\Omega\times\mathbb{N})}.$$

More generally the Clark formula implies the following.

Corollary 8.6 *Let $a \in \mathbb{N}$ and $F \in L^2(\Omega)$. We have*

$$F = \mathbb{E}[F \mid \mathcal{F}_a] + \sum_{k=a+1}^{\infty}\mathbb{E}[D_kF \mid \mathcal{F}_{k-1}]Y_k, \tag{8.7}$$

and

$$\mathbb{E}[F^2] = \mathbb{E}[(\mathbb{E}[F \mid \mathcal{F}_a])^2] + \mathbb{E}\left[\sum_{k=a+1}^{\infty}(\mathbb{E}[D_kF \mid \mathcal{F}_{k-1}])^2\right]. \tag{8.8}$$

Proof. From Proposition 3.5 and the Clark formula (8.2) of Proposition 8.1 we have

$$\mathbb{E}[F \mid \mathcal{F}_a] = \mathbb{E}[F] + \sum_{k=0}^{a}\mathbb{E}[D_kF \mid \mathcal{F}_{k-1}]Y_k,$$

which implies (8.7). Relation (8.8) is an immediate consequence of (8.7) and the isometry property of J. □

As an application of the Clark formula of Corollary 8.6 we obtain the following predictable representation property for discrete-time martingales.

Proposition 8.9 *Let $(M_n)_{n\in\mathbb{N}}$ be a martingale in $L^2(\Omega)$ with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}}$. There exists a predictable process $(u_k)_{k\in\mathbb{N}}$ locally in $L^2(\Omega\times\mathbb{N})$, (i.e. $u(\cdot)\mathbf{1}_{[0,N]}(\cdot) \in L^2(\Omega\times\mathbb{N})$ for all $N > 0$) such that*

$$M_n = M_{-1} + \sum_{k=0}^{n} u_kY_k, \qquad n \in \mathbb{N}. \tag{8.10}$$

Proof. Let $k \geq 1$. From Corollaries 7.6 and 8.6 we have:

$$\begin{aligned} M_k &= \mathbb{E}[M_k \mid \mathcal{F}_{k-1}] + \mathbb{E}[D_kM_k \mid \mathcal{F}_{k-1}]Y_k \\ &= M_{k-1} + \mathbb{E}[D_kM_k \mid \mathcal{F}_{k-1}]Y_k, \end{aligned}$$

hence it suffices to let

$$u_k = \mathbb{E}[D_k M_k \mid \mathcal{F}_{k-1}], \qquad k \geq 0,$$

to obtain

$$M_n = M_{-1} + \sum_{k=0}^{n} M_k - M_{k-1} = M_{-1} + \sum_{k=0}^{n} u_k Y_k.$$

□

9 Divergence Operator

The divergence operator δ is introduced as the adjoint of D. Let $\mathcal{U} \subset L^2(\Omega \times \mathbb{N})$ be the space of processes defined as

$$\mathcal{U} = \left\{ \sum_{k=0}^{n} J_k(f_{k+1}(*,\cdot)), \quad f_{k+1} \in \ell^2(\mathbb{N})^{\circ k} \otimes \ell^2(\mathbb{N}), \quad k =, n \in \mathbb{N} \right\}.$$

Definition 9.1 *Let $\delta : \mathcal{U} \to L^2(\Omega)$ be the linear mapping defined on $\mathcal{U}$ as*

$$\delta(u) = \delta(J_n(f_{n+1}(*,\cdot))) = J_{n+1}(\tilde{f}_{n+1}), \quad f_{n+1} \in l^2(\mathbb{N})^{\circ n} \otimes l^2(\mathbb{N}),$$

for $(u_k)_{k\in\mathbb{N}}$ of the form

$$u_k = J_n(f_{n+1}(*,k)), \qquad k \in \mathbb{N},$$

where $\tilde{f}_{n+1}$ denotes the symmetrization of f_{n+1} in $n+1$ variables, i.e.

$$\tilde{f}_{n+1}(k_1,\ldots,k_{n+1}) = \frac{1}{n+1} \sum_{i=1}^{n+1} f_{n+1}(k_1,\ldots,k_{k-1},k_{k+1},\ldots,k_{n+1},k_i).$$

From Proposition 6.7, $\mathcal{S}$ is dense in $L^2(\Omega)$, hence $\mathcal{U}$ is dense in $L^2(\Omega \times \mathbb{N})$.

Proposition 9.2 *The operator δ is adjoint to D:*

$$\mathbb{E}[\langle DF, u \rangle_{\ell^2(\mathbb{N})}] = \mathbb{E}[F\delta(u)], \quad F \in \mathcal{S},\ u \in \mathcal{U}.$$

Proof. We consider $F = J_n(f_n)$ and $u_k = J_m(g_{m+1}(*,k))$, $k \in \mathbb{N}$, where $f_n \in \ell^2(\mathbb{N})^{\circ n}$ and $g_{m+1} \in \ell^2(\mathbb{N})^{\circ m} \otimes \ell^2(\mathbb{N})$. We have

$$\begin{aligned}\mathbb{E}[\langle D.J_n(f_n), J_m(g_{m+1}(*,\cdot))\rangle_{\ell^2(\mathbb{N})}] &= n\mathbb{E}[\langle J_{n-1}(f_n(*,\cdot)), J_m(g_m(*,\cdot))\rangle_{l^2(\mathbb{N})}] \\ &= n\mathbf{1}_{\{n-1=m\}} \sum_{k=0}^{\infty} \mathbb{E}[J_{n-1}(f_n(*,k)\mathbf{1}_{\Delta_n}(*,k))J_m(g_{m+1}(*,k))]\end{aligned}$$

$$
\begin{aligned}
&= n!\mathbf{1}_{\{n-1=m\}} \sum_{k=0}^{\infty} \langle \mathbf{1}_{\Delta_n}(*,k) f_n(*,k), g_{m+1}(*,k) \rangle_{\ell^2(\mathbb{N}^{n-1})} \\
&= n!\mathbf{1}_{\{n=m+1\}} \langle \mathbf{1}_{\Delta_n} f_n, g_{m+1} \rangle_{\ell^2(\mathbb{N}^n)} \\
&= n!\mathbf{1}_{\{n=m+1\}} \langle \mathbf{1}_{\Delta_n} f_n, \tilde{g}_{m+1} \rangle_{\ell^2(\mathbb{N}^n)} \\
&= \mathbb{E}[J_n(f_n) J_m(\tilde{g}_{m+1})] \\
&= \mathbb{E}[\delta(u) F].
\end{aligned}
$$

□

The next proposition shows in particular that δ coincides with the stochastic integral operator J on the square-summable predictable processes.

Proposition 9.3 *The operator δ can be extended to $u \in L^2(\Omega \times \mathbb{N})$ with*

$$
\delta(u) = \sum_{k=0}^{\infty} u_k Y_k - \sum_{k=0}^{\infty} D_k u_k - \delta(\varphi D u), \tag{9.4}
$$

provided that all series converges in $L^2(\Omega)$, where $(\varphi_k)_{k \in \mathbb{N}}$ appears in the structure equation (5.1). We also have for all $u \in \mathcal{U}$:

$$
\mathbb{E}[|\delta(u)|^2] = \mathbb{E}[\|u\|^2_{\ell^2(\mathbb{N})}] + \mathbb{E}\left[\sum_{\substack{k,l=0 \\ k \neq l}}^{\infty} D_k u_l D_l u_k - \sum_{k=0}^{\infty} (D_k u_k)^2 \right]. \tag{9.5}
$$

Proof. Using the expression (4.5) of $u_k = J_n(f_{n+1}(*,k))$ we have

$$
\begin{aligned}
\delta(u) &= J_{n+1}(\tilde{f}_{n+1}) \\
&= \sum_{(i_1,\ldots,i_{n+1}) \in \Delta_{n+1}} \tilde{f}_{n+1}(i_1,\ldots,i_{n+1}) Y_{i_1} \cdots Y_{i_{n+1}} \\
&= \sum_{k=0}^{\infty} \sum_{(i_1,\ldots,i_n) \in \Delta_n} \tilde{f}_{n+1}(i_1,\ldots,i_n,k) Y_{i_1} \cdots Y_{i_n} Y_k \\
&\quad - n \sum_{k=0}^{\infty} \sum_{(i_1,\ldots,i_{n-1}) \in \Delta_{n-1}} \tilde{f}_{n+1}(i_1,\ldots,i_{n-1},k,k) Y_{i_1} \cdots Y_{i_{n-1}} |Y_k|^2 \\
&= \sum_{k=0}^{\infty} u_k Y_k - \sum_{k=0}^{\infty} D_k u_k |Y_k|^2 \\
&= \sum_{k=0}^{\infty} u_k Y_k - \sum_{k=0}^{\infty} D_k u_k - \sum_{k=0}^{\infty} \varphi_k D_k u_k Y_k.
\end{aligned}
$$

Next, we note the commutation relation[1]

$$\begin{aligned}
D_k\delta(u) &= D_k\left(\sum_{l=0}^{\infty} u_l Y_l - \sum_{l=0}^{\infty} |Y_l|^2 D_l u_l\right)\\
&= \sum_{l=0}^{\infty}\left(Y_l D_k u_l + u_l D_k Y_l - \frac{X_k}{\sqrt{p_k q_k}} D_k u_l D_k Y_l\right)\\
&\quad -\sum_{l=0}^{\infty}\left(|Y_l|^2 D_k D_l u_l + D_l u_l D_k|Y_l|^2 - \frac{X_k}{\sqrt{p_k q_k}} D_k|Y_l|^2 D_k D_l u_l\right)\\
&= \delta(D_k u) + u_k D_k Y_k - \frac{X_k}{\sqrt{p_k q_k}} D_k u_k D_k Y_k - D_k u_k D_k|Y_k|^2\\
&= \delta(D_k u) + u_k - \left(\frac{X_k}{\sqrt{p_k q_k}} + 2Y_k D_k Y_k - \frac{X_k}{\sqrt{p_k q_k}} D_k Y_k D_k Y_k\right) D_k u_k\\
&= \delta(D_k u) + u_k - 2Y_k D_k u_k.
\end{aligned}$$

On the other hand, we have

$$\begin{aligned}
\delta(\mathbf{1}_{\{k\}} D_k u_k) &= \sum_{l=0}^{\infty} Y_l \mathbf{1}_{\{k\}}(l) D_k u_k - \sum_{l=0}^{\infty} |Y_l|^2 D_l(\mathbf{1}_{\{k\}}(l) D_k u_k)\\
&= Y_k D_k u_k - |Y_k|^2 D_k D_k u_k\\
&= Y_k D_k u_k,
\end{aligned}$$

hence

$$\begin{aligned}
\|\delta(u)\|^2_{L^2(\Omega)} &= \mathbb{E}[\langle u, D\delta(u)\rangle_{\ell^2(\mathbb{N})}]\\
&= \mathbb{E}\left[\sum_{k=0}^{\infty} u_k(u_k + \delta(D_k u) - 2Y_k D_k u_k)\right]\\
&= \mathbb{E}[\|u\|^2_{\ell^2(\mathbb{N})}] + \mathbb{E}\left[\sum_{k,l=0}^{\infty} D_k u_l D_l u_k\right] - 2\mathbb{E}\left[\sum_{k=0}^{\infty} u_k Y_k D_k u_k\right]\\
&= \mathbb{E}[\|u\|^2_{\ell^2(\mathbb{N})}] + \mathbb{E}\left[\sum_{k,l=0}^{\infty} D_k u_l D_l u_k - 2\sum_{k=0}^{\infty}(D_k u_k)^2\right],
\end{aligned}$$

where we used the equality

$$\begin{aligned}
\mathbb{E}\left[u_k Y_k D_k u_k\right] &= \mathbb{E}\left[p_k \mathbf{1}_{\{X_k=1\}} u_k(\omega_+^k) Y_k(\omega_+^k) D_k u_k + q_k \mathbf{1}_{\{X_k=-1\}} u_k(\omega_-^k) Y_k(\omega_-^k) D_k u_k\right]\\
&= \sqrt{p_k q_k}\,\mathbb{E}\left[(\mathbf{1}_{\{X_k=1\}} u_k(\omega_+^k) - \mathbf{1}_{\{X_k=-1\}} u_k(\omega_-^k)) D_k u_k\right]\\
&= \mathbb{E}\left[(D_k u_k)^2\right], \qquad k \in \mathbb{N}.
\end{aligned}$$

□

[1] See A. Mantei, Masterarbeit "Stochastisches Kalkül in diskreter Zeit", Satz 6.7, 2015.

In the symmetric case $p_k = q_k = 1/2$ we have $\varphi_k = 0$, $k \in \mathbb{N}$, and

$$\delta(u) = \sum_{k=0}^{\infty} u_k Y_k - \sum_{k=0}^{\infty} D_k u_k.$$

The last two terms in the right hand side of (9.4) vanish when $(u_k)_{k\in\mathbb{N}}$ is predictable, and in this case the Skorohod isometry (9.5) becomes the Itô isometry as in the next proposition.

Corollary 9.6 *If $(u_k)_{k\in\mathbb{N}}$ satisfies $D_k u_k = 0$, i.e. u_k does not depend on X_k, $k \in \mathbb{N}$, then $\delta(u)$ coincides with the (discrete time) stochastic integral*

$$\delta(u) = \sum_{k=0}^{\infty} Y_k u_k, \tag{9.7}$$

provided that the series converges in $L^2(\Omega)$. If moreover $(u_k)_{k\in\mathbb{N}}$ is predictable and square-summable we have the isometry

$$\mathbb{E}[\delta(u)^2] = \mathbb{E}\left[\|u\|^2_{\ell^2(\mathbb{N})}\right], \tag{9.8}$$

and $\delta(u)$ coincides with $J(u)$ on the space of predictable square-summable processes.

10 Ornstein-Uhlenbeck Semi-Group and Process

The Ornstein-Uhlenbeck operator L is defined as $L = \delta D$, i.e. L satisfies

$$LJ_n(f_n) = nJ_n(f_n), \qquad f_n \in \ell^2(\mathbb{N})^{\circ n}.$$

Proposition 10.1 *For any $F \in \mathcal{S}$ we have*

$$LF = \delta DF = \sum_{k=0}^{\infty} Y_k (D_k F) = \sum_{k=0}^{\infty} \sqrt{p_k q_k} Y_k (F_k^+ - F_k^-),$$

Proof. Note that $D_k D_k F = 0$, $k \in \mathbb{N}$, and use Relation (9.4) of Proposition 9.3. □

Note that L can be expressed in other forms, for example

$$LF = \sum_{k=0}^{\infty} \Delta_k F,$$

where

$$\Delta_k F \;=\; (\mathbf{1}_{\{X_k=1\}} q_k (F(\omega) - F(\omega_-^k)) - \mathbf{1}_{\{X_k=-1\}} p_k (F(\omega_+^k) - F(\omega)))$$

$$
\begin{aligned}
&= F - (\mathbf{1}_{\{X_k=1\}} q_k F(\omega_-^k) + \mathbf{1}_{\{X_k=-1\}} p_k F(\omega_+^k)) \\
&= F - \mathbb{E}[F \mid \mathcal{F}_k^c], \quad k \in \mathbb{N},
\end{aligned}
$$

and $\mathcal{F}_k^c$ is the σ-algebra generated by

$$
\{X_l \ : \ l \neq k, \ l \in \mathbb{N}\}.
$$

Let now $(P_t)_{t\in\mathbb{R}_+} = (e^{tL})_{t\in\mathbb{R}_+}$ denote the semi-group associated to L and defined as

$$
P_t F = \sum_{n=0}^{\infty} e^{-nt} J_n(f_n), \qquad t \in \mathbb{R}_+,
$$

on $F = \sum_{n=0}^{\infty} J_n(f_n) \in L^2(\Omega)$. The next result shows that $(P_t)_{t\in\mathbb{R}_+}$ admits an integral representation by a probability kernel. Let $q_t^N : \Omega \times \Omega \to \mathbb{R}_+$ be defined by

$$
q_t^N(\tilde{\omega}, \omega) = \prod_{i=0}^{N} (1 + e^{-t} Y_i(\omega) Y_i(\tilde{\omega})), \qquad \omega, \tilde{\omega} \in \Omega, \quad t \in \mathbb{R}_+.
$$

Lemma 10.2 *Let the probability kernel $Q_t(\tilde{\omega}, d\omega)$ be defined by*

$$
\mathbb{E}\left[\frac{dQ_t(\tilde{\omega}, \cdot)}{d\mathbb{P}} \Big| \mathcal{F}_N \right](\omega) = q_t^N(\tilde{\omega}, \omega), \qquad N \geq 1, \quad t \in \mathbb{R}_+.
$$

For $F \in L^2(\Omega, \mathcal{F}_N)$ we have

$$
P_t F(\tilde{\omega}) = \int_\Omega F(\omega) Q_t(\tilde{\omega}, d\omega), \qquad \tilde{\omega} \in \Omega, \quad n \geq N. \tag{10.3}
$$

Proof. Since $L^2(\Omega, \mathcal{F}_N)$ has finite dimension 2^{N+1}, it suffices to consider functionals of the form $F = Y_{k_1} \cdots Y_{k_n}$ with $0 \leq k_1 < \cdots < k_n \leq N$. We have for $\omega \in \Omega$, $k \in \mathbb{N}$:

$$
\begin{aligned}
&\mathbb{E}\left[Y_k(\cdot)(1 + e^{-t} Y_k(\cdot) Y_k(\omega)) \right] \\
&\quad = p_k \sqrt{\frac{q_k}{p_k}} \left(1 + e^{-t} \sqrt{\frac{q_k}{p_k}} Y_k(\omega) \right) - q_k \sqrt{\frac{p_k}{q_k}} \left(1 - e^{-t} \sqrt{\frac{p_k}{q_k}} Y_k(\omega) \right) \\
&\quad = e^{-t} Y_k(\omega),
\end{aligned}
$$

which implies, by independence of the sequence $(X_k)_{k\in\mathbb{N}}$,

$$
\begin{aligned}
\mathbb{E}[Y_{k_1} \cdots Y_{k_n} q_t^N(\omega, \cdot)] &= \mathbb{E}\left[Y_{k_1} \cdots Y_{k_n} \prod_{i=1}^{N} (1 + e^{-t} Y_{k_i}(\omega) Y_{k_i}(\cdot)) \right] \\
&= \prod_{i=1}^{N} \mathbb{E}\left[Y_{k_i}(\cdot)(1 + e^{-t} Y_{k_i}(\omega) Y_{k_i}(\cdot)) \right]
\end{aligned}
$$

$$
\begin{aligned}
&= e^{-nt} Y_{k_1}(\omega) \cdots Y_{k_n}(\omega) \\
&= e^{-nt} J_n(\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_n)\}})(\omega) \\
&= P_t J_n(\tilde{\mathbf{1}}_{\{(k_1,\ldots,k_n)\}})(\omega) \\
&= P_t(Y_{k_1} \cdots Y_{k_n})(\omega).
\end{aligned}
$$

□

Consider the Ω-valued stationary process $(X(t))_{t\in\mathbb{R}_+} = ((X_k(t))_{k\in\mathbb{N}})_{t\in\mathbb{R}_+}$ with independent components and distribution given by

$$\mathbb{P}(X_k(t) = 1 \mid X_k(0) = 1) = p_k + e^{-t} q_k, \tag{10.4}$$

$$\mathbb{P}(X_k(t) = -1 \mid X_k(0) = 1) = q_k - e^{-t} q_k, \tag{10.5}$$

$$\mathbb{P}(X_k(t) = 1 \mid X_k(0) = -1) = p_k - e^{-t} p_k, \tag{10.6}$$

$$\mathbb{P}(X_k(t) = -1 \mid X_k(0) = -1) = q_k + e^{-t} p_k, \tag{10.7}$$

$k \in \mathbb{N}$, $t \in \mathbb{R}_+$.

Proposition 10.8 *The process $(X(t))_{t\in\mathbb{R}_+} = ((X_k(t))_{k\in\mathbb{N}})_{t\in\mathbb{R}_+}$ is the Ornstein-Uhlenbeck process associated to $(P_t)_{t\in\mathbb{R}_+}$, i.e. we have*

$$P_t F = \mathbb{E}[F(X(t)) \mid X(0)], \qquad t \in \mathbb{R}_+. \tag{10.9}$$

Proof. By construction of $(X(t))_{t\in\mathbb{R}_+}$ in Relations (10.4)-(10.7) we have

$$\mathbb{P}(X_k(t) = 1 \mid X_k(0)) = p_k \left(1 + e^{-t} Y_k \sqrt{\frac{q_k}{p_k}}\right),$$

$$\mathbb{P}(X_k(t) = -1 \mid X_k(0)) = q_k \left(1 - e^{-t} Y_k \sqrt{\frac{p_k}{q_k}}\right),$$

$$\mathbb{P}(X_k(t) = 1 \mid X_k(0)) = p_k \left(1 + e^{-t} Y_k \sqrt{\frac{q_k}{p_k}}\right),$$

$$\mathbb{P}(X_k(t) = -1 \mid X_k(0)) = q_k \left(1 - e^{-t} Y_k \sqrt{\frac{p_k}{q_k}}\right),$$

thus

$$d\mathbb{P}(X_k(t)(\tilde{\omega}) = \epsilon \mid X(0))(\omega) = \left(1 + e^{-t} Y_k(\omega) Y_k(\tilde{\omega})\right) d\mathbb{P}(X_k(\tilde{\omega}) = \epsilon),$$

$\varepsilon = \pm 1$. Since the components of $(X_k(t))_{k \in \mathbb{N}}$ are independent, this shows that the law of $(X_0(t), \ldots, X_n(t))$ conditionally to $X(0)$ has the density $q_t^n(\tilde{\omega}, \cdot)$ with respect to $\mathbb{P}$:

$$\begin{aligned} & d\mathbb{P}(X_0(t)(\tilde{\omega}) = \epsilon_0, \ldots, X_n(t)(\tilde{\omega}) = \epsilon_n \mid X(0))(\tilde{\omega}) \\ & \quad = \quad q_t^n(\tilde{\omega}, \omega) d\mathbb{P}(X_0(\tilde{\omega}) = \epsilon_0, \ldots, X_n(\tilde{\omega}) = \epsilon_n). \end{aligned}$$

Consequently we have

$$\mathbb{E}[F(X(t)) \mid X(0) = \tilde{\omega}] = \int_\Omega F(\omega) q_t^N(\tilde{\omega}, \omega) \mathbb{P}(d\omega), \tag{10.10}$$

hence from (10.3), Relation (10.9) holds for $F \in L^2(\Omega, \mathcal{F}_N)$, $N \geq 0$. □

The independent components $X_k(t)$, $k \in \mathbb{N}$, can be constructed from the data of $X_k(0) = \epsilon$ and an independent exponential random variable τ_k via the following procedure. If $\tau_k > t$, let $X_k(t) = X_k(0) = \epsilon$, otherwise if $\tau_k < t$, take $X_k(t)$ to be an independent copy of X_k. This procedure is illustrated in the following equalities:

$$\begin{aligned} \mathbb{P}(X_k(t) = 1 \mid X_k(0) = 1) &= \mathbb{E}[\mathbf{1}_{\{\tau_k > t\}}] + \mathbb{E}[\mathbf{1}_{\{\tau_k < t\}} \mathbf{1}_{\{X_k = 1\}}] \\ &= e^{-t} + p_k(1 - e^{-t}), \end{aligned} \tag{10.11}$$

$$\begin{aligned} \mathbb{P}(X_k(t) = -1 \mid X_k(0) = 1) &= \mathbb{E}[\mathbf{1}_{\{\tau_k < t\}} \mathbf{1}_{\{X_k = -1\}}] \\ &= q_k(1 - e^{-t}), \end{aligned} \tag{10.12}$$

$$\begin{aligned} \mathbb{P}(X_k(t) = -1 \mid X_k(0) = -1) &= \mathbb{E}[\mathbf{1}_{\{\tau_k > t\}}] + \mathbb{E}[\mathbf{1}_{\{\tau_k < t\}} \mathbf{1}_{\{X_k = -1\}}] \\ &= e^{-t} + q_k(1 - e^{-t}), \end{aligned} \tag{10.13}$$

$$\mathbb{P}(X_k(t) = 1 \mid X_k(0) = -1) = \mathbb{E}[\mathbf{1}_{\{\tau_k < t\}} \mathbf{1}_{\{X_k = 1\}}] \tag{10.14}$$

$$= \quad p_k(1-e^{-t}).$$

The operator $L^2(\Omega\times\mathbb{N}) \to L^2(\Omega\times\mathbb{N})$ which maps $(u_k)_{k\in\mathbb{N}}$ to $(P_t u_k)_{k\in\mathbb{N}}$ is also denoted by P_t. As a consequence of the representation of P_t given in Lemma 10.2 we obtain the following bound.

Lemma 10.15 *For $F \in \mathrm{Dom}\,(D)$ we have*

$$\|P_t u\|_{L^\infty(\Omega,\ell^2(\mathbb{N}))} \leq \|u\|_{L^\infty(\Omega,\ell^2(\mathbb{N}))}, \qquad t\in\mathbb{R}_+, \quad u\in L^2(\Omega\times\mathbb{N}).$$

Proof. As a consequence of the representation formula (10.10) we have $\mathbb{P}(d\tilde{\omega})$-a.s.:

$$\begin{aligned}
\|P_t u\|^2_{\ell^2(\mathbb{N})}(\tilde{\omega}) &= \sum_{k=0}^\infty |P_t u_k(\tilde{\omega})|^2 \\
&= \sum_{k=0}^\infty \left(\int_\Omega u_k(\omega) Q_t(\tilde{\omega}, d\omega)\right)^2 \\
&\leq \sum_{k=0}^\infty \int_\Omega |u_k(\omega)|^2 Q_t(\tilde{\omega}, d\omega) \\
&= \int_\Omega \|u\|^2_{\ell^2(\mathbb{N})}(\omega) Q_t(\tilde{\omega}, d\omega) \\
&\leq \|u\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}.
\end{aligned}$$

□

11 Covariance Identities

In this section we state the covariance identities which will be used for the proof of deviation inequalities in the next section. The covariance $\mathrm{Cov}(F,G)$ of $F,G\in L^2(\Omega)$ is defined as

$$\mathrm{Cov}(F,G) = \mathbb{E}[(F-\mathbb{E}[F])(G-\mathbb{E}[G])] = \mathbb{E}[FG] - \mathbb{E}[F]\mathbb{E}[G].$$

Proposition 11.1 *We have for $F,G\in L^2(\Omega)$ such that $\mathbb{E}[\|DF\|^2_{\ell^2(\mathbb{N})}]<\infty$:*

$$\mathrm{Cov}(F,G) = \mathbb{E}\left[\sum_{k=0}^\infty \mathbb{E}\left[D_k G \mid \mathcal{F}_{k-1}\right] D_k F\right]. \tag{11.2}$$

Proof. This identity is a consequence of the Clark formula (8.2):

$$\mathrm{Cov}(F,G) \quad = \quad \mathbb{E}[(F-\mathbb{E}[F])(G-\mathbb{E}[G])]$$

$$
\begin{aligned}
&= \mathbb{E}\left[\left(\sum_{k=0}^{\infty}\mathbb{E}[D_kF \mid \mathcal{F}_{k-1}]Y_k\right)\left(\sum_{l=0}^{\infty}\mathbb{E}[D_lG \mid \mathcal{F}_{l-1}]Y_l\right)\right]\\
&= \mathbb{E}\left[\sum_{k=0}^{\infty}\mathbb{E}[D_kF \mid \mathcal{F}_{k-1}]\mathbb{E}[D_kG \mid \mathcal{F}_{k-1}]\right]\\
&= \sum_{k=0}^{\infty}\mathbb{E}\left[\mathbb{E}[\mathbb{E}[D_kG \mid \mathcal{F}_{k-1}]D_kF \mid \mathcal{F}_{k-1}]\right]\\
&= \mathbb{E}\left[\sum_{k=0}^{\infty}\mathbb{E}[D_kG \mid \mathcal{F}_{k-1}]D_kF\right].
\end{aligned}
$$

□

A covariance identity can also be obtained using the semi-group $(P_t)_{t\in\mathbb{R}_+}$.

Proposition 11.3 *For any $F, G \in L^2(\Omega)$ such that*

$$
\mathbb{E}[\|DF\|^2_{\ell^2(\mathbb{N})}] < \infty \qquad \text{and} \qquad \mathbb{E}[\|DG\|^2_{\ell^2(\mathbb{N})}] < \infty,
$$

we have

$$
\mathrm{Cov}(F,G) = \mathbb{E}\left[\sum_{k=0}^{\infty}\int_0^{\infty} e^{-t}(D_kF)P_tD_kG dt\right]. \tag{11.4}
$$

Proof. Consider $F = J_n(f_n)$ and $G = J_m(g_m)$. We have

$$
\begin{aligned}
&\mathrm{Cov}(J_n(f_n), J_m(g_m)) = \mathbb{E}\left[J_n(f_n)J_m(g_m)\right]\\
&= \mathbf{1}_{\{n=m\}}n!\langle f_n, g_n\mathbf{1}_{\Delta_n}\rangle_{\ell^2(\mathbb{N}^n)}\\
&= \mathbf{1}_{\{n=m\}}n!n\int_0^{\infty} e^{-nt}dt\langle f_n, g_n\mathbf{1}_{\Delta_n}\rangle_{\ell^2(\mathbb{N}^n)}\\
&= \mathbf{1}_{\{n-1=m-1\}}n!n\int_0^{\infty} e^{-t}\sum_{k=0}^{\infty}\langle f_n(*,k), e^{-(n-1)t}g_n(*,k)\mathbf{1}_{\Delta_n}(*,k)\rangle_{\ell^2(\mathbb{N}^{n-1})}dt\\
&= nm\mathbb{E}\left[\int_0^{\infty} e^{-t}\sum_{k=0}^{\infty}J_{n-1}(f_n(*,k)\mathbf{1}_{\Delta_n}(*,k))e^{-(m-1)t}J_{m-1}(g_m(*,k)\mathbf{1}_{\Delta_m}(*,k))dt\right]\\
&= nm\mathbb{E}\left[\int_0^{\infty} e^{-t}\sum_{k=0}^{\infty}J_{n-1}(f_n(*,k)\mathbf{1}_{\Delta_n}(*,k))P_tJ_{m-1}(g_m(*,k)\mathbf{1}_{\Delta_m}(*,k))dt\right]\\
&= \mathbb{E}\left[\int_0^{\infty} e^{-t}\sum_{k=0}^{\infty}D_kJ_n(f_n)P_tD_kJ_m(g_m)dt\right].
\end{aligned}
$$

□

From (10.11)-(10.14) the covariance identity (11.4) shows that

$$\begin{aligned}\mathrm{Cov}(F,G) &= \mathbb{E}\left[\sum_{k=0}^{\infty}\int_0^{\infty} e^{-t} D_k F P_t D_k G dt\right]\\ &= \mathbb{E}\left[\int_0^1 \sum_{k=0}^{\infty} D_k F P_{-\log\alpha} D_k G d\alpha\right]\\ &= \int_0^1 \int_{\Omega\times\Omega} \sum_{k=0}^{\infty} D_k F(\omega) D_k G((\omega_i \mathbf{1}_{\{\tau_i<-\log\alpha\}} + \omega_i' \mathbf{1}_{\{\tau_i<-\log\alpha\}})_{i\in\mathbb{N}}) d\alpha \mathbb{P}(d\omega)\mathbb{P}(d\omega')\\ &= \int_0^1 \int_{\Omega\times\Omega} \sum_{k=0}^{\infty} D_k F(\omega) D_k G((\omega_i \mathbf{1}_{\{\xi_i<\alpha\}} + \omega_i' \mathbf{1}_{\{\xi_i>\alpha\}})_{i\in\mathbb{N}}) \mathbb{P}(d\omega)\mathbb{P}(d\omega') d\alpha,\end{aligned} \tag{11.5}$$

where $(\xi_i)_{i\in\mathbb{N}}$ is a family of i.i.d. random variables, uniformly distributed on $[0,1]$. Note that the marginals of $(X_k, X_k \mathbf{1}_{\{\xi_k<\alpha\}} + X_k' \mathbf{1}_{\{\xi_i>\alpha\}})$ are identical when X_k' is an independent copy of X_k. Let

$$\phi_\alpha(s,t) = \mathbb{E}[e^{isX_k} e^{it(X_k + \mathbf{1}_{\{\xi_k<\alpha\}}) + it(X_k' + \mathbf{1}_{\{\xi_k>\alpha\}})}].$$

Then we have the relation

$$\phi_\alpha(s,t) = \alpha\phi(s+t) + (1-\alpha)\phi(s)\phi(t), \quad \alpha\in[0,1].$$

Note that

$$\mathrm{Cov}(e^{isX_k}, e^{itX_k}) = \phi_1(s,t) - \phi_0(s,t) = \int_0^1 \frac{d\phi_\alpha}{d\alpha}(s,t) d\alpha = \phi(s+t) - \phi(s)\phi(t).$$

Next we prove an iterated version of the covariance identity in discrete time, which is an analog of a result proved in [15] for the Wiener and Poisson processes.

Theorem 11.6 *Let $n\in\mathbb{N}$ and $F,G\in L^2(\Omega)$. We have*

$$\begin{aligned}\mathrm{Cov}(F,G) = \sum_{d=1}^{d=n} (-1)^{d+1} \mathbb{E}\left[\sum_{\{1\le k_1<\cdots<k_d\}} (D_{k_d}\cdots D_{k_1}F)(D_{k_d}\cdots D_{k_1}G)\right] \\ + (-1)^n \mathbb{E}\left[\sum_{\{1\le k_1<\cdots<k_{n+1}\}} (D_{k_{n+1}}\cdots D_{k_1}F)\mathbb{E}\left[D_{k_{n+1}}\cdots D_{k_1}G \mid \mathcal{F}_{k_{n+1}-1}\right]\right].\end{aligned} \tag{11.7}$$

Proof. Take $F=G$. For $n=0$, (11.7) is a consequence of the Clark formula. Let $n\ge 1$. Applying Lemma 8.6 to $D_{k_n}\cdots D_{k_1}F$ with $a=k_n$ and $b=k_{n+1}$, and

summing on $(k_1,\ldots,k_n)\in\Delta_n$, we obtain

$$\mathbb{E}\left[\sum_{\{1\leq k_1<\cdots<k_n\}}\left(\mathbb{E}[D_{k_n}\cdots D_{k_1}F\mid\mathcal{F}_{k_n-1}]\right)^2\right]=\mathbb{E}\left[\sum_{\{1\leq k_1<\cdots<k_n\}}\mid D_{k_n}\cdots D_{k_1}F|^2\right]$$
$$-\mathbb{E}\left[\sum_{\{1\leq k_1<\cdots<k_{n+1}\}}\left(\mathbb{E}\left[D_{k_{n+1}}\cdots D_{k_1}F\mid\mathcal{F}_{k_{n+1}-1}\right]\right)^2\right],$$

which concludes the proof by induction and bilinearity. □

As a consequence of Theorem 11.6, letting $F=G$ we get the variance inequality

$$\sum_{k=1}^{2n}\frac{(-1)^{k+1}}{k!}\mathbb{E}\left[\|D^kF\|^2_{\ell^2(\Delta_k)}\right]\leq\mathrm{Var}(F)\leq\sum_{k=1}^{2n-1}\frac{(-1)^{k+1}}{k!}\mathbb{E}\left[\|D^kF\|^2_{\ell^2(\Delta_k)}\right],$$

since

$$\begin{aligned}
&\mathbb{E}\left[\sum_{\{1\leq k_1<\cdots<k_{n+1}\}}(D_{k_{n+1}}\cdots D_{k_1}F)\mathbb{E}\left[D_{k_{n+1}}\cdots D_{k_1}G\mid\mathcal{F}_{k_{n+1}-1}\right]\right]\\
&=\mathbb{E}\left[\sum_{\{1\leq k_1<\cdots<k_{n+1}\}}\mathbb{E}\left[(D_{k_{n+1}}\cdots D_{k_1}F)\mathbb{E}\left[D_{k_{n+1}}\cdots D_{k_1}G\mid\mathcal{F}_{k_{n+1}-1}\right]\mid\mathcal{F}_{k_{n+1}-1}\right]\right]\\
&=\mathbb{E}\left[\sum_{\{1\leq k_1<\cdots<k_{n+1}\}}(\mathbb{E}\left[D_{k_{n+1}}\cdots D_{k_1}G\mid\mathcal{F}_{k_{n+1}-1}\right])^2\right]\\
&\geq 0,
\end{aligned}$$

see Relation (2.15) in [15] in continuous time. In a similar way, another iterated covariance identity can be obtained from Proposition 11.3.

Corollary 11.8 *Let $n\in\mathbb{N}$ and $F,G\in L^2(\Omega,\mathcal{F}_N)$. We have*

$$\begin{aligned}
\mathrm{Cov}(F,G)&=\sum_{d=1}^{d=n}(-1)^{d+1}\mathbb{E}\left[\sum_{\{1\leq k_1<\cdots<k_d\leq N\}}(D_{k_d}\cdots D_{k_1}F)(D_{k_d}\cdots D_{k_1}G)\right]\\
&+(-1)^n\int_{\Omega\times\Omega}\sum_{\{1\leq k_1<\cdots<k_{n+1}\leq N\}}D_{k_{n+1}}\cdots D_{k_1}F(\omega)D_{k_{n+1}}\cdots D_{k_1}G(\omega')\\
&\qquad q_t^N(\omega,\omega')\mathbb{P}(d\omega)\mathbb{P}(d\omega').
\end{aligned}\tag{11.9}$$

The covariance and variance have the tensorization property:

$$\mathrm{Var}(FG)=\mathbb{E}[F\,\mathrm{Var}\,G]+\mathbb{E}[G\,\mathrm{Var}\,F]$$

if F, G are independent, hence most of the identities in this section can be obtained by tensorization of a one dimensional elementary covariance identity.

An elementary consequence of the covariance identities is the following lemma.

Lemma 11.10 *Let $F, G \in L^2(\Omega)$ such that*

$$\mathbb{E}[D_k F|\mathcal{F}_{k-1}] \cdot \mathbb{E}[D_k G|\mathcal{F}_{k-1}] \geq 0, \qquad k \in \mathbb{N}.$$

Then F and G are non-negatively correlated:

$$\mathrm{Cov}(F, G) \geq 0.$$

According to the next definition, a non-decreasing functional F satisfies $D_k F \geq 0$ for all $k \in \mathbb{N}$.

Definition 11.11 *A random variable $F : \Omega \to \mathbb{R}$ is said to be non-decreasing if for all $\omega_1, \omega_2 \in \Omega$ we have*

$$\omega_1(k) \leq \omega_2(k), \qquad k \in \mathbb{N}, \qquad \Rightarrow \ F(\omega_1) \leq F(\omega_2).$$

The following result is then immediate from Proposition 7.3 and Lemma 11.10, and shows that the FKG inequality holds on Ω. It can also be obtained from from Proposition 11.3.

Proposition 11.12 *If $F, G \in L^2(\Omega)$ are non-decreasing then F and G are non-negatively correlated:*

$$\mathrm{Cov}(F, G) \geq 0.$$

Note however that the assumptions of Lemma 11.10 are actually weaker as they do not require F and G to be non-decreasing.

12 Deviation Inequalities

In this section, which is based on [16], we recover a deviation inequality of [5] in the case of Bernoulli measures, using covariance representations instead of the logarithmic Sobolev inequalities to be presented in Section 13. The method relies on a bound on the Laplace transform $L(t) = \mathbb{E}[e^{tF}]$ obtained via a differential inequality and Chebychev's inequality.

Proposition 12.1 *Let $F:\Omega\to\mathbb{R}$ be such that $|F_k^+ - F_k^-|\leq K$, $k\in\mathbb{N}$, for some $K\geq 0$, and $\|DF\|_{L^\infty(\Omega,\ell^2(\mathbb{N}))}<\infty$. Then*

$$\begin{aligned}\mathbb{P}(F-\mathbb{E}[F]\geq x) &\leq \exp\left(-\frac{\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}}{K^2}g\left(\frac{xK}{\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}}\right)\right)\\ &\leq \exp\left(-\frac{x}{2K}\log\left(1+\frac{xK}{\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}}\right)\right),\end{aligned}$$

with $g(u)=(1+u)\log(1+u)-u$, $u\geq 0$.

Proof. Although D_k does not satisfy a derivation rule for products, from Proposition 7.8 we have

$$\begin{aligned}D_k e^F &= \mathbf{1}_{\{X_k=1\}}\sqrt{p_kq_k}(e^F-e^{F_k^-})+\mathbf{1}_{\{X_k=-1\}}\sqrt{p_kq_k}(e^{F_k^+}-e^F)\\ &= \mathbf{1}_{\{X_k=1\}}\sqrt{p_kq_k}e^F(1-e^{-\frac{1}{\sqrt{p_kq_k}}D_kF})+\mathbf{1}_{\{X_k=-1\}}\sqrt{p_kq_k}e^F(e^{\frac{1}{\sqrt{p_kq_k}}D_kF}-1)\\ &= -X_k\sqrt{p_kq_k}e^F(e^{-\frac{X_k}{\sqrt{p_kq_k}}D_kF}-1),\end{aligned}$$

hence

$$D_k e^F = X_k\sqrt{p_kq_k}e^F(1-e^{-\frac{X_k}{\sqrt{p_kq_k}}D_kF}), \tag{12.2}$$

and since the function $x\mapsto(e^x-1)/x$ is positive and increasing on $\mathbb{R}$ we have:

$$\frac{e^{-sF}D_ke^{sF}}{D_kF} = -\frac{X_k\sqrt{p_kq_k}}{D_kF}\left(e^{-s\frac{X_k}{\sqrt{p_kq_k}}D_kF}-1\right)\leq\frac{e^{sK}-1}{K},$$

or in other terms:

$$\frac{e^{-sF}D_ke^{sF}}{D_kF} = \mathbf{1}_{\{X_k=1\}}\frac{e^{s(F_k^--F_k^+)}-1}{F_k^--F_k^+}+\mathbf{1}_{\{X_k=-1\}}\frac{e^{s(F_k^+-F_k^-)}-1}{F_k^+-F_k^-}\leq\frac{e^{sK}-1}{K}.$$

We first assume that F is a bounded random variable with $\mathbb{E}[F]=0$. From Lemma 10.15 applied to DF, we have

$$\begin{aligned}\mathbb{E}[Fe^{sF}] &= \mathrm{Cov}(F,e^{sF})\\ &= \mathbb{E}\left[\int_0^\infty e^{-v}\sum_{k=0}^\infty D_ke^{sF}P_vD_kFdv\right]\\ &\leq \left\|\frac{e^{-sF}De^{sF}}{DF}\right\|_\infty\mathbb{E}\left[e^{sF}\int_0^\infty e^{-v}\|DFP_vDF\|_{\ell^1(\mathbb{N})}dv\right]\\ &\leq \frac{e^{sK}-1}{K}\mathbb{E}\left[e^{sF}\|DF\|_{\ell^2(\mathbb{N})}\int_0^\infty e^{-v}\|P_vDF\|_{\ell^2(\mathbb{N})}dv\right]\\ &\leq \frac{e^{sK}-1}{K}\mathbb{E}\left[e^{sF}\right]\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}\int_0^\infty e^{-v}dv\end{aligned}$$

$$\leq \frac{e^{sK}-1}{K}\mathbb{E}\left[e^{sF}\right]\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}.$$

In the general case, letting $L(s)=\mathbb{E}[e^{s(F-\mathbb{E}[F])}]$, we have

$$\begin{aligned}\log(\mathbb{E}[e^{t(F-\mathbb{E}[F])}]) &= \int_0^t \frac{L'(s)}{L(s)}ds\\ &= \int_0^t \frac{\mathbb{E}[(F-\mathbb{E}[F])e^{s(F-\mathbb{E}[F])}]}{\mathbb{E}[e^{s(F-\mathbb{E}[F])}]}ds\\ &\leq \frac{1}{K}\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}\int_0^t(e^{sK}-1)ds\\ &= \frac{1}{K^2}(e^{tK}-tK-1)\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))},\end{aligned}$$

$t\geq 0$. We have for all $x\geq 0$ and $t\geq 0$:

$$\begin{aligned}\mathbb{P}(F-\mathbb{E}[F]\geq x) &\leq e^{-tx}\mathbb{E}[e^{t(F-\mathbb{E}[F])}]\\ &\leq \exp\left(\frac{1}{K^2}(e^{tK}-tK-1)\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}-tx\right),\end{aligned}$$

The minimum in $t\geq 0$ in the above expression is attained with

$$t=\frac{1}{K}\log\left(1+\frac{xK}{\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}}\right),$$

hence

$$\begin{aligned}&\mathbb{P}(F-\mathbb{E}[F]\geq x)\\ &\leq \exp\left(-\frac{1}{K}\left(\left(x+\frac{1}{K}\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}\right)\log\left(1+xK\|DF\|^{-2}_{L^\infty(\Omega,\ell^2(\mathbb{N}))}\right)-x\right)\right)\\ &\leq \exp\left(-\frac{x}{2K}\log\left(1+xK\|DF\|^{-2}_{L^\infty(\Omega,\ell^2(\mathbb{N}))}\right)\right),\end{aligned}$$

where we used the inequality $(1+u)\log(1+u)-u\geq\frac{u}{2}\log(1+u)$. If $K=0$, the above proof is still valid by replacing all terms by their limits as $K\to 0$. If F is not bounded the conclusion holds for $F_n=\max(-n,\min(F,n))$, $n\geq 1$, and $(F_n)_{n\in\mathbb{N}}$, $(DF_n)_{n\in\mathbb{N}}$, converge respectively almost surely and in $L^2(\Omega\times\mathbb{N})$ to F and DF, with $\|DF_n\|^2_{L^\infty(\Omega,L^2(\mathbb{N}))}\leq\|DF\|^2_{L^\infty(\Omega,L^2(\mathbb{N}))}$. □

In case $p_k=p$ for all $k\in\mathbb{N}$, the conditions

$$\frac{1}{\sqrt{pq}}|D_kF|\leq\beta,\quad k\in\mathbb{N},\quad\text{and}\quad\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}\leq\alpha^2,$$

give

$$\mathbb{P}(F-\mathbb{E}[F]\geq x)\leq\exp\left(-\frac{\alpha^2pq}{\beta^2}g\left(\frac{x\beta}{\alpha^2pq}\right)\right)\leq\exp\left(-\frac{x}{2\beta}\log\left(1+\frac{x\beta}{\alpha^2pq}\right)\right),$$

which is Relation (13) in [5]. In particular if F is $\mathcal{F}_N$-measurable, then

$$\mathbb{P}(F-\mathbb{E}[F]\geq x)\leq \exp\left(-Ng\left(\frac{x}{\beta N}\right)\right)\leq \exp\left(-\frac{x}{\beta}\left(\log\left(1+\frac{x}{\beta N}\right)-1\right)\right).$$

Finally we show a Gaussian concentration inequality for functionals of $(S_n)_{n\in\mathbb{N}}$, using the covariance identity (11.2). We refer to [3], [4], [17], [20], for other versions of this inequality.

Proposition 12.3 *Let $F:\Omega\to\mathbb{R}$ be such that*

$$\left\|\sum_{k=0}^{\infty}\frac{1}{2(p_k\wedge q_k)}|D_kF|\|D_kF\|_{\infty}\right\|_{\infty}\leq K^2.$$

Then

$$\mathbb{P}(F-\mathbb{E}[F]\geq x)\leq \exp\left(-\frac{x^2}{2K^2}\right),\qquad x\geq 0. \tag{12.4}$$

Proof. Again, we assume that F is a bounded random variable with $\mathbb{E}[F]=0$. Using the inequality

$$|e^{tx}-e^{ty}|\leq\frac{t}{2}|x-y|(e^{tx}+e^{ty}),\qquad x,y\in\mathbb{R}, \tag{12.5}$$

we have

$$\begin{aligned}
|D_ke^{tF}| &= \sqrt{p_kq_k}|e^{tF_k^+}-e^{tF_k^-}|\\
&\leq \frac{1}{2}\sqrt{p_kq_k}t|F_k^+-F_k^-|(e^{tF_k^+}+e^{tF_k^-})\\
&= \frac{1}{2}t|D_kF|(e^{tF_k^+}+e^{tF_k^-})\\
&\leq \frac{t}{2(p_k\wedge q_k)}|D_kF|\mathbb{E}\left[e^{tF}\mid X_i,\ i\neq k\right] \qquad (12.6)\\
&= \frac{1}{2(p_k\wedge q_k)}t\mathbb{E}\left[e^{tF}|D_kF|\mid X_i,\ i\neq k\right],
\end{aligned}$$

where in (12.6) the inequality is due to the absence of chain rule of derivation for the operator D_k. Now, Proposition 11.1 yields

$$\begin{aligned}
\mathbb{E}[Fe^{tF}] &= \mathrm{Cov}(F,e^{sF})\\
&= \sum_{k=0}^{\infty}\mathbb{E}[\mathbb{E}[D_kF\mid\mathcal{F}_{k-1}]D_ke^{tF}]\\
&\leq \sum_{k=0}^{\infty}\|D_kF\|_{\infty}\mathbb{E}\left[|D_ke^{tF}|\right]
\end{aligned}$$

$$
\begin{aligned}
&\leq \frac{t}{2}\sum_{k=0}^{\infty}\frac{1}{p_k\wedge q_k}\|D_kF\|_\infty\mathbb{E}\left[\mathbb{E}\left[e^{tF}|D_kF|\mid X_i,\ i\neq k\right]\right]\\
&= \frac{t}{2}\mathbb{E}\left[e^{tF}\sum_{k=0}^{\infty}\frac{1}{p_k\wedge q_k}\|D_kF\|_\infty|D_kF|\right]\\
&\leq \frac{t}{2}\mathbb{E}[e^{tF}]\left\|\sum_{k=0}^{\infty}\frac{1}{p_k\wedge q_k}|D_kF|\|D_kF\|_\infty\right\|_\infty.
\end{aligned}
$$

This shows that

$$
\begin{aligned}
\log(\mathbb{E}[e^{t(F-\mathbb{E}[F])}]) &= \int_0^t\frac{\mathbb{E}[(F-\mathbb{E}[F])e^{s(F-\mathbb{E}[F])}]}{\mathbb{E}[e^{s(F-\mathbb{E}[F])}]}ds\\
&\leq K^2\int_0^t sds\\
&= \frac{t^2}{2}K^2,
\end{aligned}
$$

hence

$$
\begin{aligned}
e^x\mathbb{P}(F-\mathbb{E}[F]\geq x) &\leq \mathbb{E}[e^{t(F-\mathbb{E}[F])}]\\
&\leq e^{t^2K^2/2},\qquad t\geq 0,
\end{aligned}
$$

and

$$
\mathbb{P}(F-\mathbb{E}[F]\geq x)\leq e^{\frac{t^2}{2}K^2-tx},\quad t\geq 0.
$$

The best inequality is obtained for $t=x/K^2$. If F is not bounded the conclusion holds for $F_n=\max(-n,\min(F,n))$, $n\geq 0$, and $(F_n)_{n\in\mathbb{N}}$, $(DF_n)_{n\in\mathbb{N}}$, converge respectively to F and DF in $L^2(\Omega)$, resp. $L^2(\Omega\times\mathbb{N})$, with $\|DF_n\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}\leq\|DF\|^2_{L^\infty(\Omega,\ell^2(\mathbb{N}))}$. □

The bound (12.4) implies $\mathbb{E}[e^{\alpha|F|}]<\infty$ for all $\alpha>0$, and $\mathbb{E}[e^{\alpha F^2}]<\infty$ for all $\alpha<1/(2K^2)$. In case $p_k=p$, $k\in\mathbb{N}$, we obtain

$$
\mathbb{P}(F-\mathbb{E}[F]\geq x)\leq\exp\left(-\frac{px^2}{\|DF\|^2_{\ell^2(\mathbb{N},L^\infty(\Omega))}}\right).
$$

13 Logarithmic Sobolev Inequalities

The logarithmic Sobolev inequalities on Gaussian space provide an infinite dimensional analog of Sobolev inequalities, cf. e.g. [21]. On Riemannian path space [6] and on Poisson space [1], [28], martingale methods have been successfully applied to

the proof of logarithmic Sobolev inequalities. Here, discrete time martingale methods are used along with the Clark predictable representation formula (8.2) as in [10], to provide a proof of logarithmic Sobolev inequalities for Bernoulli measures. Here we are only concerned with modified logarithmic Sobolev inequalities, and we refer to [25], Theorem 2.2.8 and references therein, for the standard version of the logarithmic Sobolev inequality on the hypercube under Bernoulli measures.

The entropy of a random variable $F > 0$ is defined by

$$\mathrm{Ent}\,[F] = \mathbb{E}[F \log F] - \mathbb{E}[F] \log \mathbb{E}[F],$$

for sufficiently integrable F.

Lemma 13.1 *The entropy has the tensorization property, i.e. if F, G are sufficiently integrable independent random variables we have*

$$\mathrm{Ent}\,[FG] = \mathbb{E}[F\mathrm{Ent}\,[G]] + \mathbb{E}[G\mathrm{Ent}\,[F]]. \tag{13.2}$$

Proof. We have

$$\begin{aligned}
\mathrm{Ent}\,[FG] &= \mathbb{E}[FG \log(FG)] - \mathbb{E}[FG] \log \mathbb{E}[FG] \\
&= \mathbb{E}[FG(\log F + \log G)] - \mathbb{E}[F]\mathbb{E}[G](\log \mathbb{E}[F] + \log \mathbb{E}[G]) \\
&= \mathbb{E}[G]\mathbb{E}[F \log F] + \mathbb{E}[F]\mathbb{E}[G \log G)] - \mathbb{E}[F]\mathbb{E}[G](\log \mathbb{E}[F] + \log \mathbb{E}[G]) \\
&= \mathbb{E}[F\mathrm{Ent}\,[G]] + \mathbb{E}[G\mathrm{Ent}\,[F]].
\end{aligned}$$

□

In the next proposition we recover the modified logarithmic Sobolev inequality of [5] using the Clark representation formula in discrete time.

Theorem 13.3 *Let $F \in \mathrm{Dom}\,(D)$ with $F > \eta$ a.s. for some $\eta > 0$. We have*

$$\mathrm{Ent}\,[F] \leq \mathbb{E}\left[\frac{1}{F}\|DF\|^2_{\ell^2(\mathbb{N})}\right]. \tag{13.4}$$

Proof. Assume that F is $\mathcal{F}_N$-measurable and let $M_n = \mathbb{E}[F \mid \mathcal{F}_n]$, $0 \leq n \leq N$. Using Corollary 7.6 and the Clark formula (8.2) we have

$$M_n = M_{-1} + \sum_{k=0}^{n} u_k Y_k, \qquad 0 \leq n \leq N,$$

with $u_k = \mathbb{E}[D_k F \mid \mathcal{F}_{k-1}]$, $0 \leq k \leq n \leq N$, and $M_{-1} = \mathbb{E}[F]$. Letting $f(x) = x \log x$ and using the bound

$$\begin{aligned} f(x+y) - f(x) &= y \log x + (x+y) \log \left(1 + \frac{y}{x}\right) \\ &\leq y(1 + \log x) + \frac{y^2}{x}, \end{aligned}$$

we have:

$$\begin{aligned} \mathrm{Ent}\,[F] &= \mathbb{E}[f(M_N)] - \mathbb{E}[f(M_{-1})] \\ &= \mathbb{E}\left[\sum_{k=0}^{N} f(M_k) - f(M_{k-1})\right] \\ &= \mathbb{E}\left[\sum_{k=0}^{N} f\left(M_{k-1} + Y_k u_k\right) - f(M_{k-1})\right] \\ &\leq \mathbb{E}\left[\sum_{k=0}^{N} Y_k u_k (1 + \log M_{k-1}) + \frac{Y_k^2 u_k^2}{M_{k-1}}\right] \\ &= \mathbb{E}\left[\sum_{k=0}^{N} \frac{1}{\mathbb{E}[F \mid \mathcal{F}_{k-1}]} (\mathbb{E}[D_k F \mid \mathcal{F}_{k-1}])^2\right] \\ &\leq \mathbb{E}\left[\sum_{k=0}^{N} \mathbb{E}\left[\frac{1}{F} |D_k F|^2 \mid \mathcal{F}_{k-1}\right]\right] \\ &= \mathbb{E}\left[\frac{1}{F} \sum_{k=0}^{N} |D_k F|^2\right]. \end{aligned}$$

where we used the Jensen inequality and the convexity of $(u, v) \mapsto v^2/u$ on $(0, \infty) \times \mathbb{R}$, or the Schwarz inequality applied to $1/\sqrt{F}$ and $(D_k F/\sqrt{F})_{k \in \mathbb{N}}$, as in the Wiener and Poisson cases [6] and [1]. This inequality is extended by density to $F \in \mathrm{Dom}\,(D)$.

□

Theorem 13.3 can also be recovered by the tensorization Lemma 13.1 and the following one-variable argument: letting $p + q = 1$, $p, q > 0$, $f : \{-1, 1\} \to (0, \infty)$, $\mathbb{E}[f] = pf(1) + qf(-1)$, and $\mathrm{d}f = f(1) - f(-1)$ we have:

$$\begin{aligned} &\mathrm{Ent}\,[f] = pf(1) \log f(1) + qf(-1) \log f(-1) - \mathbb{E}[f] \log \mathbb{E}[f] \\ &= pf(1) \log(\mathbb{E}[f] + q\mathrm{d}f) + qf(-1) \log(\mathbb{E}[f] - p\mathrm{d}f) - (pf(1) + qf(-1)) \log \mathbb{E}[f] \\ &= pf(1) \log \left(1 + q \frac{\mathrm{d}f}{\mathbb{E}[f]}\right) + qf(-1) \log \left(1 - p \frac{\mathrm{d}f}{\mathbb{E}[f]}\right) \\ &\leq pqf(1) \frac{\mathrm{d}f}{\mathbb{E}[f]} - pqf(-1) \frac{\mathrm{d}f}{\mathbb{E}[f]} = pq \frac{|\mathrm{d}f|^2}{\mathbb{E}[f]} \end{aligned}$$

$$\leq \quad pq\mathbb{E}\left[\frac{1}{f}|\mathrm{d}f|^2\right].$$

Similarly we have

$$\begin{aligned}
\mathrm{Ent}\,[f] &= pf(1)\log f(1) + qf(-1)\log f(-1) - \mathbb{E}[f]\log\mathbb{E}[f] \\
&= p(\mathbb{E}[f] + q\mathrm{d}f)\log(\mathbb{E}[f] + q\mathrm{d}f) \\
&\quad +q(\mathbb{E}[f] - p\mathrm{d}f)\log(\mathbb{E}[f] - p\mathrm{d}f) - (pf(1) + qf(-1))\log\mathbb{E}[f] \\
&= p\mathbb{E}[f]\log\left(1 + q\frac{\mathrm{d}f}{\mathbb{E}[f]}\right) + pq\mathrm{d}f\log f(1) \\
&\quad +q\mathbb{E}[f]\log\left(1 - p\frac{\mathrm{d}f}{\mathbb{E}[f]}\right) - qp\mathrm{d}f\log f(-1) \\
&\leq pq\mathrm{d}f\log f(1) - pq\mathrm{d}f\log f(-1) \\
&= pq\mathbb{E}\left[\mathrm{d}f\mathrm{d}\log f\right],
\end{aligned}$$

which, by tensorization, recovers the following L^1 inequality of [11], [7], and proved in [28] in the Poisson case. In the next proposition we state and prove this inequality in the multidimensional case, using the Clark representation formula, similarly to Theorem 13.3.

Theorem 13.5 *Let $F > 0$ be $\mathcal{F}_N$-measurable. We have*

$$\mathrm{Ent}\,[F] \leq \mathbb{E}\left[\sum_{k=0}^{N} D_k F D_k \log F\right]. \tag{13.6}$$

Proof. Let $f(x) = x\log x$ and

$$\Psi(x,y) = (x+y)\log(x+y) - x\log x - (1+\log x)y, \quad x,\ x+y > 0.$$

From the relation

$$\begin{aligned}
Y_k u_k &= Y_k\mathbb{E}[D_k F \mid \mathcal{F}_{k-1}] \\
&= q_k\mathbf{1}_{\{X_k=1\}}\mathbb{E}[(F_k^+ - F_k^-) \mid \mathcal{F}_{k-1}] + p_k\mathbf{1}_{\{X_k=-1\}}\mathbb{E}[(F_k^- - F_k^+) \mid \mathcal{F}_{k-1}] \\
&= \mathbf{1}_{\{X_k=1\}}\mathbb{E}[(F_k^+ - F_k^-)\mathbf{1}_{\{X_k=-1\}} \mid \mathcal{F}_{k-1}] + \mathbf{1}_{\{X_k=-1\}}\mathbb{E}[(F_k^- - F_k^+)\mathbf{1}_{\{X_k=1\}} \mid \mathcal{F}_{k-1}],
\end{aligned}$$

we have, using the convexity of Ψ:

$$\mathrm{Ent}\,[F] = \mathbb{E}\left[\sum_{k=0}^{N} f\left(M_{k-1} + Y_k u_k\right) - f(M_{k-1})\right]$$

$$
\begin{aligned}
&= \mathbb{E}\left[\sum_{k=0}^{N} \Psi(M_{k-1}, Y_k u_k) + Y_k u_k (1 + \log M_{k-1})\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} \Psi(M_{k-1}, Y_k u_k)\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} p_k \Psi\left(\mathbb{E}[F \mid \mathcal{F}_{k-1}], \mathbb{E}[(F_k^+ - F_k^-)\mathbf{1}_{\{X_k=-1\}} \mid \mathcal{F}_{k-1}]\right)\right. \\
&\quad \left. + q_k \Psi\left(\mathbb{E}[F \mid \mathcal{F}_{k-1}], \mathbb{E}[(F_k^- - F_k^+)\mathbf{1}_{\{X_k=1\}} \mid \mathcal{F}_{k-1}]\right)\right] \\
&\leq \mathbb{E}\left[\sum_{k=0}^{N} \mathbb{E}\left[p_k \Psi\left(F, (F_k^+ - F_k^-)\mathbf{1}_{\{X_k=-1\}}\right) + q_k \Psi\left(F, (F_k^- - F_k^+)\mathbf{1}_{\{X_k=1\}}\right) \mid \mathcal{F}_{k-1}\right]\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} p_k \mathbf{1}_{\{X_k=-1\}} \Psi\left(F_k^-, F_k^+ - F_k^-\right) + q_k \mathbf{1}_{\{X_k=1\}} \Psi\left(F_k^+, F_k^- - F_k^+\right)\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} p_k q_k \Psi(F_k^-, F_k^+ - F_k^-) + p_k q_k \Psi(F_k^+, F_k^- - F_k^+)\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} p_k q_k (\log F_k^+ - \log F_k^-)(F_k^+ - F_k^-)\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} D_k F D_k \log F\right].
\end{aligned}
$$

□

The proof of Theorem 13.5 can also be obtained by first using the bound

$$
f(x+y) - f(x) = y \log x + (x+y) \log\left(1 + \frac{y}{x}\right) \leq y(1 + \log x) + y \log(x+y),
$$

and then the convexity of $(u, v) \to v(\log(u+v) - \log u)$:

$$
\begin{aligned}
\mathrm{Ent}\,[F] &= \mathbb{E}\left[\sum_{k=0}^{N} f\left(M_{k-1} + Y_k u_k\right) - f(M_{k-1})\right] \\
&\leq \mathbb{E}\left[\sum_{k=0}^{N} Y_k u_k (1 + \log M_{k-1}) + Y_k u_k \log(M_{k-1} + Y_k u_k)\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} Y_k u_k (\log(M_{k-1} + Y_k u_k) - \log M_{k-1})\right] \\
&= \mathbb{E}\left[\sum_{k=0}^{N} \sqrt{p_k q_k} \mathbb{E}[D_k F \mid \mathcal{F}_{k-1}](\log \mathbb{E}[F + (F_k^+ - F_k^-)\mathbf{1}_{\{X_k=-1\}} \mid \mathcal{F}_{k-1}] - \log \mathbb{E}[F \mid \mathcal{F}_{k-1}])\right. \\
&\quad \left. - \sqrt{p_k q_k} \mathbb{E}[D_k F \mid \mathcal{F}_{k-1}](\log \mathbb{E}[F + (F_k^- - F_k^+)\mathbf{1}_{\{X_k=-1\}} \mid \mathcal{F}_{k-1}] - \log \mathbb{E}[F \mid \mathcal{F}_{k-1}])\right]
\end{aligned}
$$

$$
\begin{aligned}
&\leq \mathbb{E}\left[\sum_{k=0}^{N}\mathbb{E}\left[\sqrt{p_kq_k}D_kF(\log(F+(F_k^+-F_k^-)\mathbf{1}_{\{X_k=-1\}})-\log F)\right.\right.\\
&\quad\left.\left.-\sqrt{p_kq_k}D_kF(\log(F+(F_k^--F_k^+)\mathbf{1}_{\{X_k=1\}})-\log F)\mid\mathcal{F}_{k-1}\right]\right]\\
&= \mathbb{E}\left[\sum_{k=0}^{N}\sqrt{p_kq_k}D_kF\mathbf{1}_{\{X_k=-1\}}(\log F_k^+-\log F_k^-)\right.\\
&\quad\left.-\sqrt{p_kq_k}D_kF\mathbf{1}_{\{X_k=1\}}(\log F_k^--\log F_k^+)\right]\\
&\leq \mathbb{E}\left[\sum_{k=0}^{N}\sqrt{p_kq_k}q_kD_kF(\log F_k^+-\log F_k^-)-\sqrt{p_kq_k}p_kD_kF(\log F_k^--\log F_k^+)\right]\\
&= \mathbb{E}\left[\sum_{k=0}^{N}D_kFD_k\log F\right].
\end{aligned}
$$

The application of Theorem 13.5 to e^F gives the following inequality for $F>0$, $\mathcal{F}_N$-measurable:

$$
\begin{aligned}
\operatorname{Ent}[e^F] &\leq \mathbb{E}\left[\sum_{k=0}^{N}D_kFD_ke^F\right]\\
&= \mathbb{E}\left[\sum_{k=0}^{N}p_kq_k\Psi(e^{F_k^-},e^{F_k^+}-e^{F_k^-})+p_kq_k\Psi(e^{F_k^+},e^{F_k^-}-e^{F_k^+})\right]\\
&= \mathbb{E}\left[\sum_{k=0}^{N}p_kq_ke^{F_k^-}((F_k^+-F_k^-)e^{F_k^+-F_k^-}-e^{F_k^+-F_k^-}+1)\right.\\
&\quad\left.+p_kq_ke^{F_k^+}((F_k^--F_k^+)e^{F_k^--F_k^+}-e^{F_k^--F_k^+}+1)\right]\\
&= \mathbb{E}\left[\sum_{k=0}^{N}p_k\mathbf{1}_{\{X_k=-1\}}e^{F_k^-}((F_k^+-F_k^-)e^{F_k^+-F_k^-}-e^{F_k^+-F_k^-}+1)\right.\\
&\quad\left.+q_k\mathbf{1}_{\{X_k=1\}}e^{F_k^+}((F_k^--F_k^+)e^{F_k^--F_k^+}-e^{F_k^--F_k^+}+1)\right]\\
&= \mathbb{E}\left[e^F\sum_{k=0}^{N}\sqrt{p_kq_k}|Y_k|(\nabla_kFe^{\nabla_kF}-e^{\nabla_kF}+1)\right]. \qquad (13.7)
\end{aligned}
$$

This implies

$$
\operatorname{Ent}[e^F]\leq\mathbb{E}\left[e^F\sum_{k=0}^{N}(\nabla_kFe^{\nabla_kF}-e^{\nabla_kF}+1)\right]. \qquad (13.8)
$$

As already noted in [7], (13.6) and the Poisson limit theorem yield the L^1 inequality of [28]. Let $M_n=(n+X_1+\cdots+X_n)/2$, $F=\varphi(M_n)$, and $p_k=\lambda/n$, $k\in\mathbb{N}$, $\lambda>0$. Then

$$
\sum_{k=0}^{n}D_kFD_k\log F
$$

$$
\begin{aligned}
&= \frac{\lambda}{n}\left(1-\frac{\lambda}{n}\right)(n-M_n)(\varphi(M_n+1)-\varphi(M_n))\log(\varphi(M_n+1)-\varphi(M_n)) \\
&\quad +\frac{\lambda}{n}\left(1-\frac{\lambda}{n}\right)M_n(\varphi(M_n)-\varphi(M_n-1))\log(\varphi(M_n)-\varphi(M_n-1)).
\end{aligned}
$$

In the limit we obtain

$$
\mathrm{Ent}\left[\varphi(U)\right] \leq \lambda\mathbb{E}[(\varphi(U+1)-\varphi(U))(\log\varphi(U+1)-\log\varphi(U))],
$$

where U is a Poisson random variable with parameter λ. In one variable we have, still letting $\mathrm{d}f = f(1)-f(-1)$,

$$
\begin{aligned}
\mathrm{Ent}\left[e^f\right] &\leq pq\mathbb{E}\left[\mathrm{d}e^f \mathrm{d}\log e^f\right] \\
&= pq(e^{f(1)}-e^{f(-1)})(f(1)-f(-1)) \\
&= pqe^{f(-1)}((f(1)-f(-1))e^{f(1)-f(-1)}-e^{f(1)-f(-1)}+1) \\
&\quad +pqe^{f(1)}((f(-1)-f(1))e^{f(-1)-f(1)}-e^{f(-1)-f(1)}+1) \\
&\leq qe^{f(-1)}((f(1)-f(-1))e^{f(1)-f(-1)}-e^{f(1)-f(-1)}+1) \\
&\quad +pe^{f(1)}((f(-1)-f(1))e^{f(-1)-f(1)}-e^{f(-1)-f(1)}+1) \\
&= \mathbb{E}[e^f(\nabla f e^{\nabla f}-e^{\nabla f}+1)],
\end{aligned}
$$

where ∇_k is the gradient operator defined in (7.7). This last inequality is not comparable to the optimal constant inequality

$$
\mathrm{Ent}\left[e^F\right] \leq \mathbb{E}\left[e^F\sum_{k=0}^{N} p_k q_k(|\nabla_k F|e^{|\nabla_k F|}-e^{|\nabla_k F|}+1)\right], \tag{13.9}
$$

of [5] since when $F_k^+ - F_k^- \geq 0$ the right-hand side of (13.9) grows as $F_k^+ e^{2F_k^+}$, instead of $F_k^+ e^{F_k^+}$ in (13.8). In fact we can prove the following inequality which improves (13.4), (13.6) and (13.9).

Theorem 13.10 *Let F be $\mathcal{F}_N$-measurable. We have*

$$
\mathrm{Ent}\left[e^F\right] \leq \mathbb{E}\left[e^F\sum_{k=0}^{N} p_k q_k(\nabla_k F e^{\nabla_k F}-e^{\nabla_k F}+1)\right]. \tag{13.11}
$$

Clearly, (13.11) is better than (13.9), (13.7) and (13.6). It also improves (13.4) from the bound

$$
xe^x - e^x + 1 \leq (e^x-1)^2, \quad x\in\mathbb{R},
$$

which implies

$$e^F(\nabla F e^{\nabla F} - e^{\nabla F} + 1) \leq e^F(e^{\nabla F} - 1)^2 = e^{-F}|\nabla e^F|^2.$$

By the tensorization property (13.2), the proof of (13.11) reduces to the following one dimensional lemma.

Lemma 13.12 *For any* $0 \leq p \leq 1$, $t \in \mathbb{R}$, $a \in \mathbb{R}$, $q = 1 - p$,

$$\begin{aligned} & pte^t + qae^a - \left(pe^t + qe^a\right) \log\left(pe^t + qe^a\right) \\ & \leq \; pq\left(qe^a\left((t-a)e^{t-a} - e^{t-a} + 1\right) + pe^t\left((a-t)e^{a-t} - e^{a-t} + 1\right)\right). \end{aligned}$$

Proof. Set

$$\begin{aligned} g(t) \;=\; & pq\left(qe^a\left((t-a)e^{t-a} - e^{t-a} + 1\right) + pe^t\left((a-t)e^{a-t} - e^{a-t} + 1\right)\right) \\ & -pte^t - qae^a + \left(pe^t + qe^a\right)\log\left(pe^t + qe^a\right). \end{aligned}$$

Then

$$g'(t) = pq\left(qe^a(t-a)e^{t-a} + pe^t\left(-e^{a-t} + 1\right)\right) - pte^t + pe^t \log(pe^t + qe^a)$$

and $g''(t) = pe^t h(t)$, where

$$h(t) = -a - 2pt - p + 2pa + p^2 t - p^2 a + \log(pe^t + qe^a) + \frac{pe^t}{pe^t + qe^a}.$$

Now,

$$\begin{aligned} h'(t) \;&=\; -2p + p^2 + \frac{2pe^t}{pe^t + qe^a} - \frac{p^2 e^{2t}}{(pe^t + qe^a)^2} \\ &=\; \frac{pq^2(e^t - e^a)(pe^t + (q+1)e^a)}{(pe^t + qe^a)^2}, \end{aligned}$$

which implies that $h'(a) = 0$, $h'(t) < 0$ for any $t < a$ and $h'(t) > 0$ for any $t > a$. Hence, for any $t \neq a$, $h(t) > h(a) = 0$, and so $g''(t) \geq 0$ for any $t \in \mathbb{R}$ and $g''(t) = 0$ if and only if $t = a$. Therefore, g' is strictly increasing. Finally, since $t = a$ is the unique root of $g' = 0$, we have that $g(t) \geq g(a) = 0$ for all $t \in \mathbb{R}$. □

This inequality improves (13.4), (13.6), and (13.9), as illustrated in one dimension in Figure 1, where the entropy is represented as a function of $p \in [0, 1]$ with $f(1) = 1$

and $f(-1) = 3.5$. The inequality (13.11) is a discrete analog of the sharp inequality on Poisson space of [28]. In the symmetric case $p_k = q_k = 1/2$, $k \in \mathbb{N}$, we have

$$\begin{aligned}
\mathrm{Ent}\left[e^F\right] &\leq \mathbb{E}\left[e^F \sum_{k=0}^{N} p_k q_k (\nabla_k F e^{\nabla_k F} - \nabla_k F + 1)\right] \\
&= \frac{1}{8}\mathbb{E}\left[\sum_{k=0}^{N} e^{F_k^-}((F_k^+ - F_k^-)e^{F_k^+ - F_k^-} - e^{F_k^+ - F_k^-} + 1)\right. \\
&\quad \left. + e^{F_k^+}((F_k^- - F_k^+)e^{F_k^- - F_k^+} - e^{F_k^- - F_k^+} + 1)\right] \\
&= \frac{1}{8}\mathbb{E}\left[\sum_{k=0}^{N} (e^{F_k^+} - e^{F_k^-})(F_k^+ - F_k^-)\right] \\
&= \frac{1}{2}\mathbb{E}\left[\sum_{k=0}^{N} D_k F D_k e^F\right],
\end{aligned}$$

which improves on (13.6).

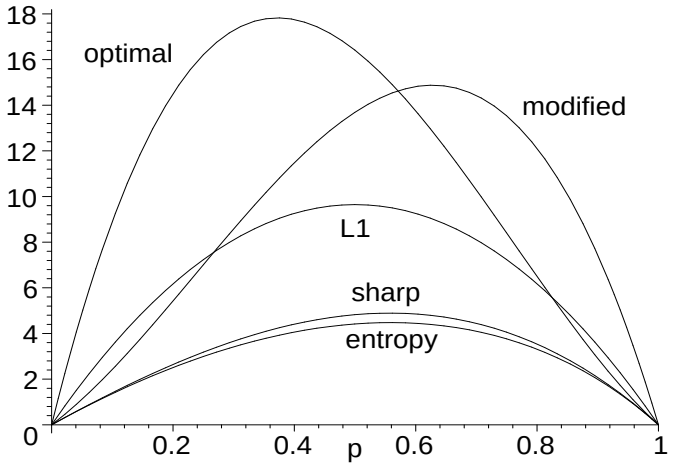


Figure 1: Graph of the entropy as a function of p.

Letting $F = \varphi(M_n)$ we have

$$\begin{aligned}
&\mathbb{E}\left[e^F \sum_{k=0}^{N} p_k q_k (\nabla_k F e^{\nabla_k F} - \nabla_k F + 1)\right] \\
&= \frac{\lambda}{n}\left(1 - \frac{\lambda}{n}\right)\mathbb{E}\left[M_n e^{\varphi(M_n)}\right. \\
&\quad \left. \times ((\varphi(M_n) - \varphi(M_n - 1))e^{\varphi(M_n) - \varphi(M_n - 1)} - e^{\varphi(M_n) - \varphi(M_n - 1)} + 1)\right]
\end{aligned}$$

$$+\frac{\lambda}{n}\left(1-\frac{\lambda}{n}\right)\mathbb{E}\left[(n-M_n)e^{\varphi(M_n)}\right.$$
$$\left.\times\left((\varphi(M_n+1)-\varphi(M_n))e^{\varphi(M_n+1)-\varphi(M_n)}-e^{\varphi(M_n+1)-\varphi(M_n)}+1\right)\right],$$

and in the limit as n goes to infinity we obtain

$$\mathrm{Ent}\left[e^{\varphi(U)}\right]\leq\lambda\mathbb{E}[e^{\varphi(U)}((\varphi(U+1)-\varphi(U))e^{\varphi(U+1)-\varphi(U)}-e^{\varphi(U+1)-\varphi(U)}+1)],$$

where U is a Poisson random variable with parameter λ. This corresponds to the sharp inequality of [28].

14 Change of Variable Formula

In this section we state a discrete-time analog of Itô's change of variable formula which will be useful for the predictable representation of random variables and for option hedging.

Proposition 14.1 *Let $(M_n)_{n\in\mathbb{N}}$ be a square-integrable martingale and $f:\mathbb{R}\times\mathbb{N}\to\mathbb{R}$. We have*

$$\begin{aligned} & f(M_n,n) \qquad (14.2)\\ &= f(M_{-1},-1)+\sum_{k=0}^{n}D_kf(M_k,k)Y_k+\sum_{k=0}^{n}\mathbb{E}[f(M_k,k)-f(M_{k-1},k-1)\mid\mathcal{F}_{k-1}].\end{aligned}$$

Proof. By Proposition 8.9 there exists square-integrable process $(u_k)_{k\in\mathbb{N}}$ such that

$$M_n=M_{-1}+\sum_{k=0}^{n}u_kY_k,\qquad n\in\mathbb{N}.$$

We write

$$\begin{aligned} f(M_n,n)-f(M_{-1},-1) &= \sum_{k=0}^{n}f(M_k,k)-f(M_{k-1},k-1)\\ &= \sum_{k=0}^{n}f(M_k,k)-f(M_{k-1},k)+f(M_{k-1},k)-f(M_{k-1},k-1)\\ &= \sum_{k=0}^{n}\sqrt{\frac{p_k}{q_k}}\left(f\left(M_{k-1}+u_k\sqrt{\frac{q_k}{p_k}},k\right)-f(M_{k-1},k)\right)Y_k\\ &\quad+\frac{p_k}{q_k}\mathbf{1}_{\{X_k=-1\}}\left(f\left(M_{k-1}+u_k\sqrt{\frac{q_k}{p_k}},k\right)-f(M_{k-1},k)\right)\\ &\quad+\mathbf{1}_{\{X_k=-1\}}\left(f\left(M_{k-1}-u_k\sqrt{\frac{p_k}{q_k}},k\right)-f(M_{k-1},k)\right)\end{aligned}$$

$$
\begin{aligned}
&+\sum_{k=0}^{n} f(M_{k-1},k)-f(M_{k-1},k-1)\\
=\;&\sum_{k=0}^{n}\sqrt{\frac{p_k}{q_k}}\left(f\left(M_{k-1}+u_k\sqrt{\frac{q_k}{p_k}},k\right)-f(M_{k-1},k)\right)Y_k\\
&+\sum_{k=0}^{n}\frac{1}{q_k}\mathbf{1}_{\{X_k=-1\}}\mathbb{E}[f(M_k,k)-f(M_{k-1},k)\mid\mathcal{F}_{k-1}]\\
&+\sum_{k=0}^{n} f(M_{k-1},k)-f(M_{k-1},k-1).
\end{aligned}
$$

Similarly we have

$$
\begin{aligned}
f(M_n,n) \;=\;& f(M_{-1},-1)-\sum_{k=0}^{n}\sqrt{\frac{q_k}{p_k}}\left(f\left(M_{k-1}-u_k\sqrt{\frac{p_k}{q_k}},k\right)-f(M_{k-1},k)\right)Y_k\\
&+\sum_{k=0}^{n}\frac{1}{p_k}\mathbf{1}_{\{X_k=1\}}\mathbb{E}[f(M_k,k)-f(M_{k-1},k)\mid\mathcal{F}_{k-1}]\\
&+\sum_{k=0}^{n} f(M_{k-1},k)-f(M_{k-1},k-1),
\end{aligned}
$$

Multiplying each increment in the above formulas respectively by q_k and p_k and summing on k we get

$$
\begin{aligned}
f(M_n,n) &= f(M_{-1},-1)\\
&+\sum_{k=0}^{n}\sqrt{p_kq_k}\left(f\left({}_{k-1}+u_k\sqrt{\frac{q_k}{p_k}},k\right)-f\left(M_{k-1}-u_k\sqrt{\frac{p_k}{q_k}},k\right)\right)Y_k\\
&+\sum_{k=0}^{n}\mathbb{E}[f(M_k,k)\mid\mathcal{F}_{k-1}]-f(M_{k-1},k).
\end{aligned}
$$

□

Note that in (14.2) we have

$$
D_kf(M_k,k)=\sqrt{p_kq_k}\left(f\left(M_{k-1}+u_k\sqrt{\frac{q_k}{p_k}},k\right)-f\left(M_{k-1}-u_k\sqrt{\frac{p_k}{q_k}},k\right)\right),\quad k\in\mathbb{N}.
$$

On the other hand, the term

$$
\mathbb{E}[f(M_k,k)-f(M_{k-1},k-1)\mid\mathcal{F}_{k-1}]
$$

is analog to the generator part in the continuous time Itô formula, and can be written as

$$
p_kf\left(M_{k-1}+u_k\sqrt{\frac{q_k}{p_k}},k\right)+q_kf\left(M_{k-1}-u_k\sqrt{\frac{p_k}{q_k}},k\right)-f\left(M_{k-1},k-1\right).
$$

When $p_n = q_n = 1/2$, $n \in \mathbb{N}$, we have

$$\begin{aligned} f(M_n, n) &= f(M_{-1}, -1) + \sum_{k=0}^{n} \frac{f\left(M_{k-1} + u_k, k\right) - f\left(M_{k-1} - u_k, k\right)}{2} Y_k \\ &\quad + \sum_{k=0}^{n} \frac{f\left(M_{k-1} + u_k, k\right) + f\left(M_{k-1} - u_k, k\right) - 2f\left(M_{k-1}, k-1\right)}{2}. \end{aligned}$$

The above proposition also provides an explicit version of the Doob decomposition for supermartingales. Naturally if $(f(M_n, n))_{n\in\mathbb{N}}$ is a martingale we have

$$\begin{aligned} f(M_n, n) &= f(M_{-1}, -1) \\ &\quad + \sum_{k=0}^{n} \sqrt{p_k q_k} \left(f\left(M_{k-1} + u_k \sqrt{\frac{q_k}{p_k}}, k \right) - f\left(M_{k-1} - u_k \sqrt{\frac{p_k}{q_k}}, k \right) \right) Y_k \\ &= f(M_{-1}, -1) + \sum_{k=0}^{n} D_k f(M_k, k) Y_k. \end{aligned}$$

In this case the Clark formula, the martingale representation formula Proposition 8.9 and the change of variable formula all coincide. In this case, we have in particular

$$D_k f(M_k, k) = \mathbb{E}[D_k f(M_n, n) \mid \mathcal{F}_{k-1}] = \mathbb{E}[D_k f(M_k, k) \mid \mathcal{F}_{k-1}], \qquad k \in \mathbb{N}.$$

If F is an $\mathcal{F}_N$-measurable random variable and f is a function such that

$$\mathbb{E}[F \mid \mathcal{F}_n] = f(M_n, n), \qquad -1 \leq n \leq N,$$

we have $F = f(M_N, N)$, $\mathbb{E}[F] = f(M_{-1}, -1)$ and

$$\begin{aligned} F &= \mathbb{E}[F] + \sum_{k=0}^{n} \mathbb{E}[D_k f(M_N, N) \mid \mathcal{F}_{k-1}] Y_k \\ &= \mathbb{E}[F] + \sum_{k=0}^{n} D_k f(M_k, k) Y_k \\ &= \mathbb{E}[F] + \sum_{k=0}^{n} D_k \mathbb{E}[f(M_N, N) \mid \mathcal{F}_k] Y_k. \end{aligned}$$

Such a function f exists if $(M_n)_{n\in\mathbb{N}}$ is Markov and $F = h(M_N)$. In this case, consider the semi-group $(P_{k,n})_{0\leq k<n\leq N}$ associated to $(M_n)_{n\in\mathbb{N}}$ and defined by

$$[P_{k,n} h](x) = \mathbb{E}[h(M_n) \mid M_k = x].$$

Letting $f(x, n) = [P_{n,N} h](x)$ we can write

$$F = \mathbb{E}[F] + \sum_{k=0}^{n} \mathbb{E}[D_k h(M_N) \mid \mathcal{F}_{k-1}] Y_k = \mathbb{E}[F] + \sum_{k=0}^{n} D_k [P_{k,N} h(M_k)] Y_k.$$

15 Option Hedging in Discrete Time

In this section we give a presentation of the Black-Scholes formula in discrete time, or Cox-Ross-Rubinstein model, see e.g. [9], [19], §15-1 of [27], or [24], as an application of the Clark formula.

In order to be consistent with the notation of the previous sections we choose to use the time scale $\mathbb{N}$, hence the index 0 is that of the first random value of any stochastic process, while the index -1 corresponds to its deterministic initial value.

Let $(A_k)_{k\in\mathbb{N}}$ be a riskless asset with initial value A_{-1}, and defined by

$$A_n = A_{-1}\prod_{k=0}^{n}(1+r_k), \qquad n\in\mathbb{N},$$

where $(r_k)_{k\in\mathbb{N}}$, is a sequence of deterministic numbers such that $r_k > -1$, $k\in\mathbb{N}$. Consider a stock price with initial value S_{-1}, given in discrete time as

$$S_n = \begin{cases} (1+b_n)S_{n-1}, & X_n = 1, \\ \\ (1+a_n)S_{n-1}, & X_n = -1, \quad n\in\mathbb{N}, \end{cases}$$

where $(a_k)_{k\in\mathbb{N}}$ and $(b_k)_{k\in\mathbb{N}}$ are sequences of deterministic numbers such that

$$-1 < a_k < r_k < b_k, \qquad k\in\mathbb{N}.$$

We have

$$S_n = S_{-1}\prod_{k=0}^{n}\sqrt{(1+b_k)(1+a_k)}\left(\frac{1+b_k}{1+a_k}\right)^{X_k/2}, \quad n\in\mathbb{N}.$$

Consider now the discounted stock price given as

$$\begin{aligned} \tilde{S}_n &= S_n\prod_{k=0}^{n}(1+r_k)^{-1} \\ &= S_{-1}\prod_{k=0}^{n}\left(\frac{1}{1+r_k}\sqrt{(1+b_k)(1+a_k)}\left(\frac{1+b_k}{1+a_k}\right)^{X_k/2}\right), \quad n\in\mathbb{N}. \end{aligned}$$

If $-1 < a_k < r_k < b_k$, $k\in\mathbb{N}$, then $(\tilde{S}_n)_{n\in\mathbb{N}}$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq -1}$ under the probability $\mathbb{P}^*$ given by

$$p_k = (r_k - a_k)/(b_k - a_k), \quad q_k = (b_k - r_k)/(b_k - a_k), \quad k\in\mathbb{N}.$$

In other terms, under $\mathbb{P}^*$ we have

$$\mathbb{E}^*[S_{n+1} \mid \mathcal{F}_n] = (1+r_{n+1})S_n, \qquad n \geq -1,$$

where $\mathbb{E}^*$ denotes the expectation under $\mathbb{P}^*$. Recall that under this probability measure there is absence of arbitrage and the market is complete. From the change of variable formula Proposition 14.1 or from the Clark formula (8.2) we have the martingale representation

$$\tilde{S}_n = S_{-1} + \sum_{k=0}^{n} Y_k D_k \tilde{S}_k = S_{-1} + \sum_{k=0}^{n} \tilde{S}_{k-1}\sqrt{p_k q_k}\frac{b_k - a_k}{1+r_k} Y_k.$$

Definition 15.1 *A portfolio strategy is a pair of predictable processes $(\eta_k)_{k\in\mathbb{N}}$ and $(\zeta_k)_{k\in\mathbb{N}}$ where η_k, resp. ζ_k represents the numbers of units invested over the time period $(k, k+1]$ in the asset S_k, resp. A_k, with $k \geq 0$.*

The value at time $k \geq -1$ of the portfolio $(\eta_k, \zeta_k)_{0\leq k\leq N}$ is defined as

$$V_k = \zeta_{k+1}A_k + \eta_{k+1}S_k, \qquad k \geq -1, \tag{15.2}$$

and its discounted value is defined as

$$\tilde{V}_n = V_n \prod_{k=0}^{n}(1+r_k)^{-1}, \qquad n \geq -1. \tag{15.3}$$

Definition 15.4 *A portfolio $(\eta_k, \zeta_k)_{k\in\mathbb{N}}$ is said to be self-financing if*

$$A_n(\zeta_{n+1} - \zeta_n) + S_n(\eta_{n+1} - \eta_n) = 0, \qquad n \geq 0.$$

Note that the self-financing condition implies

$$V_n = \zeta_n A_n + \eta_n S_n, \qquad n \geq 0.$$

Our goal is to hedge an arbitrary claim on Ω, i.e. given an $\mathcal{F}_N$-measurable random variable F we search for a portfolio $(\eta_k, \zeta_k)_{0\leq k\leq n}$ such that the equality

$$F = V_N = \zeta_N A_N + \eta_N S_N \tag{15.5}$$

holds at time $N \in \mathbb{N}$.

Proposition 15.6 *Assume that the portfolio $(\eta_k, \zeta_k)_{0\leq k\leq N}$ is self-financing. Then we have the decomposition*

$$V_n = V_{-1}\prod_{k=0}^{n}(1+r_k) + \sum_{i=0}^{n}\eta_i S_{i-1}\sqrt{p_i q_i}(b_i - a_i)Y_i \prod_{k=i+1}^{n}(1+r_k). \tag{15.7}$$

Proof. Under the self-financing assumption we have

$$\begin{aligned} V_i - V_{i-1} &= \zeta_i(A_i - A_{i-1}) + \eta_i(S_i - S_{i-1}) \\ &= r_i\zeta_i A_{i-1} + (a_i \mathbf{1}_{\{X_i=-1\}} + b_i \mathbf{1}_{\{X_i=1\}})\eta_i S_{i-1} \\ &= \eta_i S_{i-1}(a_i \mathbf{1}_{\{X_i=-1\}} + b_i \mathbf{1}_{\{X_i=1\}} - r_i) + r_i V_{i-1} \\ &= \eta_i S_{i-1}\sqrt{p_i q_i}(b_i - a_i)Y_i + r_i V_{i-1}, \qquad i \in \mathbb{N}, \end{aligned}$$

hence for the discounted portfolio we get:

$$\begin{aligned} \tilde{V}_i - \tilde{V}_{i-1} &= \prod_{k=1}^{i}(1+r_k)^{-1}V_i - \prod_{k=1}^{i-1}(1+r_k)^{-1}V_{i-1} \\ &= \prod_{k=1}^{i}(1+r_k)^{-1}(V_i - V_{i-1} - r_i V_{i-1}) \\ &= \eta_i S_{i-1}\sqrt{p_i q_i}(b_i - a_i)Y_i \prod_{k=1}^{i}(1+r_k)^{-1}, \qquad i \in \mathbb{N}, \end{aligned}$$

which successively yields (15.8) and (15.7). □

As a consequence of (15.7) and (15.3) we immediately obtain

$$\tilde{V}_n = \tilde{V}_{-1} + \sum_{i=0}^{n} \eta_i S_{i-1}\sqrt{p_i q_i}(b_i - a_i)Y_i \prod_{k=0}^{i}(1+r_k)^{-1}, \qquad n \geq -1. \tag{15.8}$$

The next proposition provides a solution to the hedging problem under the constraint (15.5).

Proposition 15.9 *Given $F \in L^2(\Omega, \mathcal{F}_N)$, let*

$$\eta_n = \frac{1}{S_{n-1}\sqrt{p_n q_n}(b_n - a_n)}\mathbb{E}^*[D_n F \mid \mathcal{F}_{n-1}] \prod_{k=n+1}^{N}(1+r_k)^{-1}, \qquad 0 \leq n \leq N, \tag{15.10}$$

and

$$\zeta_n = A_n^{-1}\left(\prod_{k=n+1}^{N}(1+r_k)^{-1}\mathbb{E}^*[F \mid \mathcal{F}_n] - \eta_n S_n\right), \qquad 0 \leq n \leq N. \tag{15.11}$$

Then the portfolio $(\eta_k, \zeta_k)_{0 \leq k \leq n}$ is self financing and satisfies

$$\zeta_n A_n + \eta_n S_n = \prod_{k=n+1}^{N}(1+r_k)^{-1}\mathbb{E}^*[F \mid \mathcal{F}_n], \qquad 0 \leq n \leq N,$$

in particular we have $V_N = F$, hence $(\eta_k, \zeta_k)_{0 \leq k \leq N}$ is a hedging strategy leading to F.

Proof. Let $(\eta_k)_{-1\le k\le N}$ be defined by (15.10) and $\eta_{-1}=0$, and consider the process $(\zeta_n)_{0\le n\le N}$ defined by

$$\zeta_{-1}=\frac{\mathbb{E}^*[F]}{S_{-1}}\prod_{k=0}^{N}(1+r_k)^{-1} \quad \text{and} \quad \zeta_{k+1}=\zeta_k-\frac{(\eta_{k+1}-\eta_k)S_k}{A_k}, \qquad k=-1,\ldots,N-1.$$

Then $(\eta_k,\zeta_k)_{-1\le k\le N}$ satisfies the self-financing condition

$$A_k(\zeta_{k+1}-\zeta_k)+S_k(\eta_{k+1}-\eta_k)=0, \qquad -1\le k\le N-1.$$

Let now

$$V_{-1}=\mathbb{E}^*[F]\prod_{k=0}^{N}(1+r_k)^{-1}, \quad \text{and} \quad V_n=\zeta_nA_n+\eta_nS_n, \qquad 0\le n\le N,$$

and

$$\tilde{V}_n=V_n\prod_{k=0}^{n}(1+r_k)^{-1}, \qquad -1\le n\le N.$$

Since $(\eta_k,\zeta_k)_{-1\le k\le N}$ is self-financing, Relation (15.8) shows that

$$\tilde{V}_n=\tilde{V}_{-1}+\sum_{i=0}^{n}Y_i\eta_iS_{i-1}\sqrt{p_iq_i}(b_i-a_i)\prod_{k=1}^{i}(1+r_k)^{-1}, \qquad -1\le n\le N. \tag{15.12}$$

On the other hand, from the Clark formula (8.2) and the definition of $(\eta_k)_{-1\le k\le N}$ we have

$$\begin{aligned}
&\mathbb{E}^*[F\mid\mathcal{F}_n]\prod_{k=0}^{N}(1+r_k)^{-1}\\
&\quad=\mathbb{E}^*\left[\mathbb{E}^*[F]\prod_{k=0}^{N}(1+r_k)^{-1}+\sum_{i=0}^{N}Y_i\mathbb{E}^*[D_iF\mid\mathcal{F}_{i-1}]\prod_{k=0}^{N}(1+r_k)^{-1}\Big|\mathcal{F}_n\right]\\
&\quad=\mathbb{E}^*[F]\prod_{k=0}^{N}(1+r_k)^{-1}+\sum_{i=0}^{n}Y_i\mathbb{E}^*[D_iF\mid\mathcal{F}_{i-1}]\prod_{k=0}^{N}(1+r_k)^{-1}\\
&\quad=\mathbb{E}^*[F]\prod_{k=0}^{N}(1+r_k)^{-1}+\sum_{i=0}^{n}Y_i\eta_iS_{i-1}\sqrt{p_iq_i}(b_i-a_i)\prod_{k=1}^{i}(1+r_k)^{-1}\\
&\quad=\tilde{V}_n
\end{aligned}$$

from (15.12). Hence

$$\tilde{V}_n=\mathbb{E}^*[F\mid\mathcal{F}_n]\prod_{k=0}^{N}(1+r_k)^{-1}, \qquad -1\le n\le N,$$

and

$$V_n = \mathbb{E}^*[F \mid \mathcal{F}_n] \prod_{k=n+1}^{N} (1+r_k)^{-1}, \qquad -1 \le n \le N.$$

In particular we have $V_N = F$. To conclude the proof we note that from the relation $V_n = \zeta_n A_n + \eta_n S_n$, $0 \le n \le N$, the process $(\zeta_n)_{0 \le n \le N}$ coincides with $(\zeta_n)_{0 \le n \le N}$ defined by (15.11). □

Note that we also have

$$\zeta_{n+1} A_n + \eta_{n+1} S_n = \mathbb{E}^*[F \mid \mathcal{F}_n] \prod_{k=n+1}^{N} (1+r_k)^{-1}, \qquad -1 \le n \le N.$$

The above proposition shows that there always exists a hedging strategy starting from

$$\tilde{V}_{-1} = \mathbb{E}^*[F] \prod_{k=0}^{N} (1+r_k)^{-1}.$$

Conversely, if there exists a hedging strategy leading to

$$\tilde{V}_N = F \prod_{k=0}^{N} (1+r_k)^{-1},$$

then $(\tilde{V}_n)_{-1 \le n \le N}$ is necessarily a martingale with initial value

$$\tilde{V}_{-1} = \mathbb{E}^*[\tilde{V}_N] = \mathbb{E}^*[F] \prod_{k=0}^{N} (1+r_k)^{-1}.$$

When $F = h(\tilde{S}_N)$, we have $\mathbb{E}^*[h(\tilde{S}_N) \mid \mathcal{F}_k] = f(\tilde{S}_k, k)$ with

$$f(x,k) = \mathbb{E}^* \left[h \left(x \prod_{i=k+1}^{n} \frac{\sqrt{(1+b_k)(1+a_k)}}{1+r_k} \left(\frac{1+b_k}{1+a_k} \right)^{X_k/2} \right) \right].$$

The hedging strategy is given by

$$\begin{aligned} \eta_k &= \frac{1}{S_{k-1}\sqrt{p_k q_k}(b_k - a_k)} D_k f(\tilde{S}_k, k) \prod_{i=k+1}^{N} (1+r_i)^{-1} \\ &= \frac{\prod_{i=k+1}^{N} (1+r_i)^{-1}}{S_{k-1}(b_k - a_k)} \left(f\left(\tilde{S}_{k-1} \frac{1+b_k}{1+r_k}, k \right) - f\left(\tilde{S}_{k-1} \frac{1+a_k}{1+r_k}, k \right) \right), \qquad k \ge -1. \end{aligned}$$

Note that η_k is non-negative (i.e. there is no short-selling) when f is a non-decreasing function, e.g. in the case of European options we have $f(x) = (x-K)^+$.

References

[1] C. Ané and M. Ledoux. On logarithmic Sobolev inequalities for continuous time random walks on graphs. *Probab. Theory Related Fields*, 116(4):573–602, 2000.

[2] S. Attal. Approximating the Fock space with the toy Fock space. In *Séminaire de Probabilités, XXXVI*, volume 1801 of *Lecture Notes in Math.*, pages 477–491. Springer, Berlin, 2003.

[3] S.G. Bobkov. On the Gross and Talagrand inequalities on the discrete cube. *Vestn. Syktyvkar. Univ. Ser. 1 Mat. Mekh. Inform.*, 1:12–19, 1995.

[4] S.G. Bobkov, C. Houdré, and P. Tetali. The subgaussian constant and concentration inequalities. *Israel J. Math.*, 156:255–283, 2006.

[5] S.G. Bobkov and M. Ledoux. On modified logarithmic Sobolev inequalities for Bernoulli and Poisson measures. *J. Funct. Anal.*, 156(2):347–365, 1998.

[6] M. Capitaine, E.P. Hsu, and M. Ledoux. Martingale representation and a simple proof of logarithmic Sobolev inequalities on path spaces. *Electron. Comm. Probab.*, 2:71–81 (electronic), 1997.

[7] P. Dai Pra, A.M. Paganoni, and G. Posta. Entropy inequalities for unbounded spin systems. *Ann. Probab.*, 30(4):1959–1976, 2002.

[8] M. Émery. A discrete approach to the chaotic representation property. In *Séminaire de Probabilités, XXXV*, volume 1755 of *Lecture Notes in Math.*, pages 123–138. Springer, Berlin, 2001.

[9] H. Föllmer and A. Schied. *Stochastic finance*, volume 27 of *de Gruyter Studies in Mathematics*. Walter de Gruyter & Co., Berlin, 2004.

[10] F.Q. Gao and N. Privault. Clark formula and logarithmic Sobolev inequalities for Bernoulli measures. *C. R. Math. Acad. Sci. Paris*, 336(1):51–56, 2003.

[11] F.Q. Gao and J. Quastel. Exponential decay of entropy in the random transposition and Bernoulli-Laplace models. *Ann. Appl. Probab.*, 13(4):1591–1600, 2003.

[12] H. Gzyl. An exposé on discrete Wiener chaos expansions. *Bol. Asoc. Mat. Venez.*, XIII(1):3–26, 2006.

[13] H. Holden, T. Lindstrøm, B. Øksendal, and J. Ubøe. Discrete Wick calculus and stochastic functional equations. *Potential Anal.*, 1(3):291–306, 1992.

[14] H. Holden, T. Lindstrøm, B. Øksendal, and J. Ubøe. Discrete Wick products. In *Stochastic analysis and related topics (Oslo, 1992)*, pages 123–148. Gordon and Breach, Montreux, 1993.

[15] C. Houdré and V. Pérez-Abreu. Covariance identities and inequalities for functionals on Wiener and Poisson spaces. *Ann. Probab.*, 23:400–419, 1995.

[16] C. Houdré and N. Privault. Concentration and deviation inequalities in infinite dimensions via covariance representations. *Bernoulli*, 8(6):697–720, 2002.

[17] C. Houdré and P. Tetali. Concentration of measure for products of Markov kernels and graph products via functional inequalities. *Combin. Probab. Comput.*, 10(1):1–28, 2001.

[18] J. Jacod and P. Protter. *Probability essentials*. Springer-Verlag, Berlin, 2000.

[19] D. Lamberton and B. Lapeyre. *Introduction to stochastic calculus applied to finance*. Chapman & Hall, London, 1996.

[20] M. Ledoux. On Talagrand's deviation inequalities for product measures. *ESAIM Probab. Statist.*, 1:63–87 (electronic), 1995/97.

[21] M. Ledoux. The geometry of Markov diffusion generators. *Ann. Fac. Sci. Toulouse Math. (6)*, 9:305–366, 2000.

[22] M. Leitz-Martini. A discrete Clark-Ocone formula. Maphysto Research Report No 29, 2000.

[23] N. Privault and W. Schoutens. Discrete chaotic calculus and covariance identities. *Stochastics and Stochastics Reports*, 72:289–315, 2002. Eurandom Report 006, 2000.

[24] J. Ruiz de Chávez. Predictable representation of the binomial process and application to options in finance. In *XXXIII National Congress of the Mexican Mathematical Society (Spanish) (Saltillo, 2000)*, volume 29 of *Aportaciones Mat. Comun.*, pages 223–230. Soc. Mat. Mexicana, México, 2001.

[25] L. Saloff-Coste. Lectures on finite Markov chains. In *Lectures on probability theory and statistics (Saint-Flour, 1996)*, volume 1665 of *Lecture Notes in Math.*, pages 301–413. Springer, Berlin, 1997.

[26] D. Stroock. Doing analysis by tossing a coin. *Math. Intelligencer*, 22(2):66–72, 2000.

[27] D. Williams. *Probability with martingales*. Cambridge Mathematical Textbooks. Cambridge University Press, Cambridge, 1991.

[28] L.M. Wu. A new modified logarithmic Sobolev inequality for Poisson point processes and several applications. *Probab. Theory Related Fields*, 118(3):427–438, 2000.